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Review Article

Computer aided diagnosis of diabetic macular edema in retinal fundus and OCT images: A review



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ARTICLE INFO

Article history:

Received 22 June 2022

Received in revised form

5 December 2022

Accepted 21 December 2022

Available online 2 January 2023

Keywords:

Diabetic macular edema

Color fundus photography

Optical coherence tomography

Deep learning

Machine learning

ABSTRACT

Diabetic Macular Edema (DME) is a potentially blinding consequence of Diabetic Retinopathy (DR) as well as the leading cause of vision loss in diabetics. DME is characterized by a buildup of extracellular fluid inside the macula through hyperpermeable vessels. The presence of DME can be spotted at any level of DR with varying degrees of severity using prominent imaging modalities such as Color Fundus Photography (CFP) and Optical Coherence Tomography (OCT). Computerized approaches for screening eye disorders appear to be beneficial, as they provide doctors with detailed insights into abnormalities. Such a system for the evaluation of retinal images can function as a stand-alone disease monitoring system. This review reports the state-of-art automated DME detection methods with traditional Machine Learning (ML) and Deep Learning (DL) techniques employing retinal fundus or OCT images. The paper provides a list of public retinal OCT and fundus imaging datasets for DME detection. In addition, the paper describes the dynamics of advancements in presented methods adopted in the past along with their strengths and limitations to highlight the insufficiencies that could be addressed in future investigations.

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<https://doi.org/10.1016/j.bbe.2022.12.005>

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1. Introduction

India is considered the “World’s Capital of Diabetes” [1]. As per the International Diabetes Federation, the incidence of diabetes has been 537 million in 2021, and it is anticipated to rise to 783 million by 2045 [2]. Through its world vision report, the World Health Organization (WHO) revealed that more than 2.2 billion individuals suffer from visual impairment or blindness, with over one billion of those having a decent likelihood of it being treated [3]. Out of these one billion individuals, 3.9 million subjects had vision-threatening DR as seen in the Fig. 1. DR is substantially correlated to a prolonged incidence of diabetes and weaker blood pressure and glycemic control. Globally, 6.96%, 6.81%, and 10.2% of the population have DR, Proliferative DR (PDR), and DME respectively. Consequently, DME impacts 21 million individuals worldwide [4]. DME is perhaps the most prevalent trigger of vision loss in DR-affected subjects [5].

DME stays the primary source of deterioration of vision in DR and it may advance next to any phase of DR progression. Detection of DME at an early stage will help sustain vision loss thereby preventing irreversible blindness. The evaluation of retinal disorders, when done manually, is often subjective, costly, and sluggish, as it depends on the experience and expertise of the ophthalmologists. With a drastic growth in the number of DME victims and the limited number of ophthalmologists, accurate and timely assessment of DME disease is difficult. Approximately 30 million OCT scans are acquired annually around the globe demanding human experts for their analysis [6].

Computer-Assisted Diagnosis Systems (CADs) have become a vital and effective tool in the clinical manifestation of ocular diseases. These systems aid in the effective therapy and quick management of patients along with support for follow-up, improving their life quality [7].

The primary objective of this review is to determine the benefits and limitations of contemporary DME classification models that have used conventional ML-based techniques as well as DL-based techniques which use CFP or OCT images in DME assessment. Furthermore, it provides a summary of the evolution of presented methods over time and empha-

sizes potential researchable issues in the context of computerized DME evaluation. For comprehensive DME assessment, the paper outlines the advantages of the DL approach and OCT imaging modality over traditional ML-based models and CFP modality. This review is intended for researchers and learners interested in using artificial intelligence in the realm of computer-based diagnosis of DME.

1.1. Domain aspects related to DME

Diabetes mellitus is a chronic health condition that is caused by altered synthesis of insulin in the blood or insulin resistance. People with diabetes suffer from excessive levels of blood sugar or glucose, resulting in hyperglycaemia which leads to long-term complications of eye, kidney, nervous system, and circulatory diseases [8]. The leading cause of blindness in working-age individuals and one of the major microvascular complications of long-standing diabetes mellitus is Diabetic Retinopathy (DR). DR is characterised by the retinal lesions known as microaneurysms, haemorrhages, hard and soft exudates, neovascularisation leading to blindness [9].

The changes in DR involve the retinal microvasculature specifically the tight junctions of the endothelial cell wall, that is, the Inner Blood Retinal Barrier (BRB). Chronic hyperglycaemia leads to the activation of numerous pathways and end products like sorbitol pathway, activated protein kinase C, advanced glycosylated end products etc. These cause capillary damage resulting in capillary leakage and obliteration. The capillary obliteration leads to hypoxia. Hypoxia in turn causes the release of vaso-formative substances which are capable of angiogenesis and the formation of new blood vessels (neovascularisation). The capillary damage may also lead to loss of pericytes (retinal endothelial supporting cells), leading to the formation of microaneurysm, increased tortuosity of retinal capillaries, plasma protein and lipid leakage [11,12]. The pathogenesis of DR has been depicted in Fig. 2. The capillary loss triggers hypoxia, which leads to an influx of growth factors that causes tissue remodelling, angiogenesis, and neovascularization. The new blood vessels can be vis-

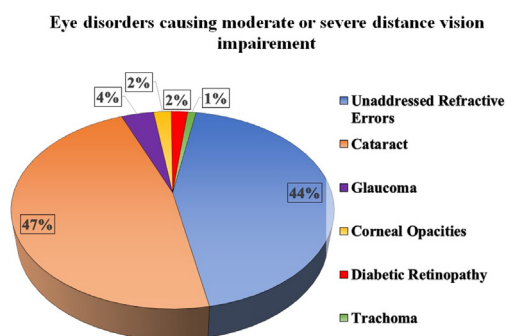


Fig. 1 – Percentage of people (in millions) suffering from various eye disorders among the one billion individuals who can be treated.

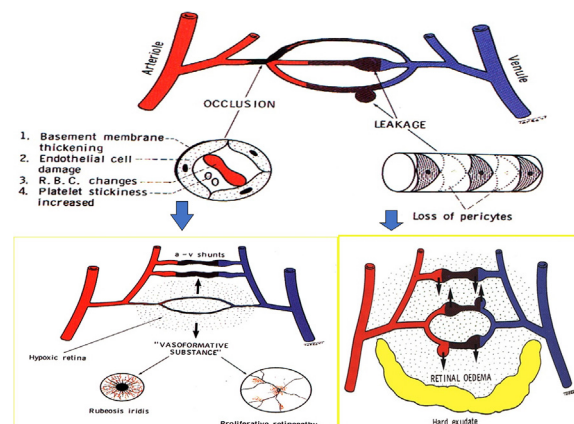


Fig. 2 – Pathogenesis of DR [10].

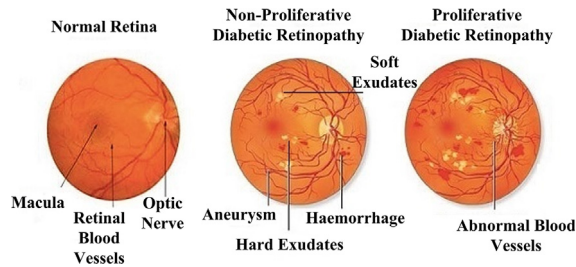


Fig. 3 – DR lesions captured using fundus camera with normal retina(Source: retinamd.com).

ible on the Optic Disc (OD) or elsewhere on the retinal surface. Fig. 3 depicts the normal fundus, non proliferative and proliferative stage of DR (showing neovascularization of the disc). These new vessels have the potential to rupture, resulting in bleeding into the vitreous cavity and tractional retinal detachment and eventually blindness.

Small protrusions or outpouchings of the retinal blood vessel wall called microaneurysms may rupture leading to bleeding in the retinal tissue along with fluid accumulation in the retina causing thickening or swelling of the macula. This thickening of the macula is termed DME or diabetic maculopathy. The thickening of the macula contributes to visual impairment in DME. DME is a complex ocular disorder that may lead to vision loss permanently [13,14].

1.2. Imaging Modalities used to screen DR and DME

In the assessment and therapy of patients with DR and DME, imaging has played a decisive role. CFP is an imaging technique that is potentially powerful tool for examining diabetic eye diseases and tracking their progression and response to treatment in people with diabetes. It helps to assess DR where lesions like microaneurysms, haemorrhages, soft and HES along with venous beading, neovascularization, and Intra Retinal Microvascular Abnormalities (IRMAs) can be visualized [14]. DR lesions are portrayed in the Fig. 3 captured using a fundus camera.

OCT is a high-resolution, non-invasive, rapid imaging style supporting a 3D view of the retina by rendering light waves. OCT is deemed to be the gold standard for managing DME [15,16]. It helps to assess retinal morphology by allowing volumetric quantification of Macular Edema (ME) along with the thickness of choroid and neurosensory retina. Fig. 4 depicts an OCT B scan of the macular region in the retina of a healthy person with twelve different retinal layers [17]. Diffuse Retinal Thickening (DRT), Cystoid Macular Edema (CME), and Serous

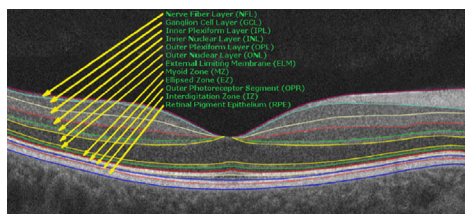


Fig. 4 – OCT Image showing 12 different retinal layers of a normal eye.

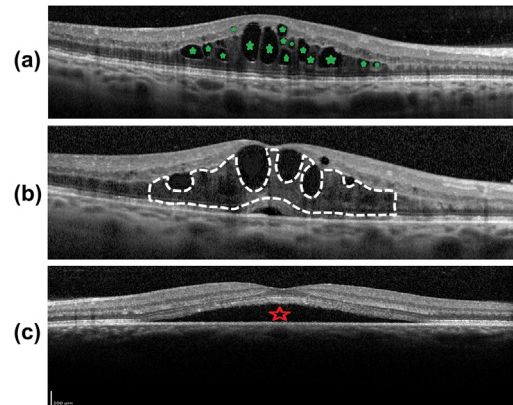


Fig. 5 – A sample OCT retinal picture showing CME: (a) (several hyporeflective cystoid spaces with star marks in green colour) [19], DRT (b) (Spongy appearance indicated by dotted lines in white) [19], and SRD (c) (a massive single hyporeflective region indicated by star mark in red) [20].

Retinal Detachment (SRD) were given as 3 different morphological forms of macular edema in medical taxonomy [18]. All these three pathologies can occur simultaneously or individually depending on the extent of the disorder. Fig. 5 displays three distinct OCT B scans with three morphological DME variants being highlighted with the regions of each type marked as described in the caption.

Public datasets are advantageous in that they are available to researchers all over the world, can be used to test and train new algorithms, as well as to compare and verify the outcomes of already developed algorithms. Tables 1 and 2 provide details of various CFP and OCT datasets that were used to train and validate state-of-the-art algorithms used in the manuscript. Engineers and researchers who are interested in investigating the domain of computerized DME detection can also benefit from the information in these Tables.

2. Literature review

An enormous amount of literature published in the past provides information on various types of methods used in the detection and classification of DME. A total of 214 publications from Scopus, IEEE Xplore, Science Direct, PubMed, Springer, Web of Science, and ARXIV are obtained through an iterative search conducted with the following keywords: Diabetic Macular Edema, DME Detection, DME Classification, DME Segmentation, Cystoid Macular Edema, CME, SRD, Serous Retinal Detachment, DRT, Diffuse Retinal Thickening, Optical Coherence Tomography, OCT, SD-OCT, Fundus, Deep Learning, DL, Machine Learning, ML, Automated DME Detection, CNN, Convolutional Neural Networks. The search was restricted to publications in science and engineering, and English language articles between the years 2010 to 2022. Few older articles and medical domain publications are included to elaborate on the basic concepts. This review covered all journal publications and conference papers along with a few book chapters that were featured in the proceedings from the above bibliographic databases. Journal articles up till October 2022 have been included. All the articles

Table 1 – Open Access fundus datasets.

SN	Dataset/Associated papers	Link to dataset	Images/File format	Device (manufacturer)	Country	Eye diseases
1	EyePACS [54]	http://www.eyepacs.com/data-analysis Accessible on registration	88702/JPEG	CR1/DGi/CR2, Optovue iCam, Topcon NW	USA	DED
2	MESSIDOR [55]	https://www.adcis.net/en/third-party/messidor/ Accessible on registration	1200/TIFF	Topcon TRC NW6 non-mydratic 3CCD camera	France	NORMAL, DED
3	MESSIDOR2 [55]56	http://www.adcis.net/en/third-party/messidor2/ Accessible on registration	1748/JPEG and PNG	Topcon TRC NW6 non-mydratic fundus camera	France	DED
4	Eophtha [57]	http://www.adcis.net/en/third-party/eophtha/ Accessible on registration	463/JPEG and PNG	NR	France	NORMAL, DED
5	HEIMED [36]58	https://github.com/IgiancaUTH/HEIMED	169/JPEG	Visucam PRO fundus camera (ZEISS)	USA	NORMAL, DED
6	DIARETDB0-Calibration-0 [59]	http://www.it.lut.fi/project/imageret/diaretdb0/	130/PNG	Digital fundus camera (Specification NR)	Finland	NORMAL, DED
7	DIARETDB1-Calibration-1 [60]	http://www.it.lut.fi/project/imageret/diaretdb1/index.html	89/PNG	Digital fundus camera (Specification NR)	Finland	NORMAL, DED
8	DRIVE	https://www.isi.uu.nl/Research/Databases/DRIVE/	40/ JPEG	Canon CR5 non-mydratic 3CCD camera	Netherlands	NORMAL, DED
9	STARE	http://cecas.clemson.edu/ahover/stare/	397	TRV-50 Topcon camera	Netherlands	NORMAL, DED
10	HRF	https://www5.cs.fau.de/research/data/fundus-images/	45/JPEG	Canon CF-60UVi camera	Germany, Czech Republic	NORMAL, DED, Glaucoma
11	IDRiD [61]	https://idrid.grand-challenge.org/Rules/	516/JPEG	Kowa VX-10 alpha digital fundus camera	India	NORMAL, DED
12	APTOS [61]	https://www.kaggle.com/c/aptos2019-blindness-detection/data	5590/PNG	Fundus camera(NR)	India	NORMAL, DED
13	ROP	http://webeye.ophth.uiowa.edu/ROC/	100/JPEG	TRC(NW200,NW100) Topcon/CR5–45NM(Canon)	Netherlands	DED
14	DDR [62]	https://github.com/nkicls/DDR-dataset	13,673/NR	42 different fundus cameras	China	DED
15	DR1	https://figshare.com/articles/Advancing_Bag_of_Visual_Words_Representations_for_Lesion_Classification_in_Retinal_Images/953671/2	1077/TIFF	TRC-50X mydratic Topcon camera	Brazil	NORMAL, DED
16	DeepDRiD	https://isbi.deepdr.org/data.html	2696/JPEG	Topcon Digital camera, Optomap P200Tx	China	NORMAL, DED
17	DR-HAGIS [63]	https://personalpages.manchester.ac.uk/staff/niall.p.mcloughlin/DRHAGIS_database.htm	40/JPEG	Topcon(TRC-NW6s,TRC-NW8), Canon CR DGi	UK	Glaucoma,AMD, Hypertension,DR

Legend: EYEPACS = Eye Picture Archive Communication System, TIFF = Tag Image File Format, JPEG = Joint Photographic Experts Group, PNG = Portable Network Graphics, AMD = Age related Macular Degeneration, DED = Diabetic Eye Disease, HEIMED = Hamilton Eye Institute ME Dataset, DIARETDB = DIAbetic RETinopathy Data Base, DRIVE = Digital Retinal Images for Vessel Extraction, STARE = STructured Analysis of Retina, HRF = High-Resolution Fundus, IDRiD = Indian DR Image Dataset, APTOS = Asia Pacific Tele-Ophthalmology Society, ROP = Retinopathy Online Challenge, DR = Diabetic Retinopathy, DDR = Dataset for DR, DeepDRiD = DeepDR DR Image Dataset, DR-HAGIS = DR,Hypertension,AMD,Glaucoma ImageS, NR = Not Reported.

Table 2 – Open Access OCT datasets.

SN	Dataset/Associated papers	Link to dataset	Diseases/Subjects/B scans	Device (manufacturer)	Country	Image format
1	BIOMISA [64][65]	https://data.mendeley.com/datasets/trghs22fpg/1/	DME, CSR, AMD, NORMAL Glaucoma/106/NR	Topcon 3-D OCT 2000	Pakistan	JPEG
2	Dukes [66]	https://people.duke.edu/~sf59/Srinivasan_BOE_2014_dataset.htm	AMD, DME, NORMAL/45/3231	Heidelberg Spectralis	USA	TIFF
3	OPTIMA Cyst segmentation challenge [67] [68]	http://optima.meduniwien.ac.at/optima-segmentation-challenge-1/	Variety of retinal cysts/30/NR Available on registration	Zeiss Cirrus, Nidek, Heidelberg Spectralis and Topcon	Austria	
4	NEH [69]	https://sites.google.com/site/hosseinrabbanikhorasgani/datasets-1	DME, AMD, NORMAL/148/4142	Heidelberg Spectralis	Iran	TIFF
5	Mendeley(Kermany dataset) [70]	https://data.mendeley.com/datasets/rscbjbr9sj/3	DME, Drusen, CNV, NORMAL/5319/ 109312	Heidelberg Spectralis	USA, China	JPEG
6	Dukes2 [71]	https://people.duke.edu/~sf59/Chiu_BOE_2014_dataset.htm	DME/ 10/ 610	Heidelberg Spectralis	USA	MAT
7	Retinal OCT Classification Challenge [58]	https://rocc.grand-challenge.org/Participation/	DR, NORMAL/ NR/ 165 Available on registration	Topcon SD-OCT device	Iran	MAT
8	RETOUCH [72]	https://retouch.grand-challenge.org	DR, NORMAL/ 112/ NR Available on registration	T1000/T2000(Topcon), Spectralis, Zeiss Cirrus	Austria, Netherlands	NR
9	OCTID [73]	http://doi.org/10.3886/E108503V1	DR,NORMAL,AMD,CSR,MH/NR/ 579 Available on registration	Cirrus HD-OCT	India	JPEG, MAT

Legend: BIOMISA = BIOMedical Image & Signal Analysis, DME = Diabetic Macular Edema, CSR = Central Serous Retinopathy, AMD = Age-related Macular Degeneration, NEH = Noor Eye Hospital, RETOUCH = The Retinal OCT Fluid Challenge, OCTID = OCT Image Retinal Database, TIFF = Tag Image File Format, JPEG = Joint Photographic Experts Group, DR = Diabetic Retinopathy, MH = Macular Hole, CNV = Choroidal Neo Vascularization, OCT = Optical Coherence Tomography, NR = Not Reported.

Table 3 – Use of fundus images with traditional ML models.

SN	Author & Year	Datasets & Images	Preprocessing Techniques	ML Techniques	Number of features	Results	Observations
1	Nayak et al. (2009) [29]	Private (350)	HE, CE	ANN	4	SE = 95.40% SP = 100%	-Manual selection of threshold value for exudate detection.
2	Siddalingaswamy & Prabhu(2010) [30]	Private (148)	GCE	Clustering, Thresholding, Morphological Reconstruction	3	SE = 95.60% SP = 96.15%	- Failed to locate macula center accurately in poor quality images.
3	Deepak et al. (2010) [21]	MESS (342)	CN, CE, BVR	CBIR, Symmetry based descriptors, LBP, IF, TF	-	Prec = 89% ACC = 95%	-Automated macula detection can be used to improve results.
4	Lim et al. (2011) [32]	MESS (88)	CLAHE, Re, GCE, IE	MP, Watershed Transforms	-	SP = 90.20% ACC = 85.20%	-Manual threshold selection. -No distinction between hard, soft exudates
5	Ang et al. (2011) [33]	Private (90)	GCE	MP, ANN	4	SP = 100% ACC = 96.67%	-Manual selection of structuring element size, threshold.
6	Hunter et al. (2011) [34]	Private (1000)	GCE	MP, MLP, Region Growing	-	SE = 97% SP = 80%	-Generalizability can be improved using different datasets
7	Akram et al. (2012) [35]	MESS (1200)	BE, ACE, NR	MP, HT, Gaf, SVM	4	SP = 97.80% ACC = 97.30%	-Magnitude of morphological closure impacts segmentation.
8	Giancardo et al. (2012) [36]	HEIMED,MESS DB	GCE, Re, BE	Exudate probability map, SVM, KNN, NB, RF	48	SVM linear kernel AUC = 0.94	-Novel features with wavelet decomposition, automatic lesion segmentation.
9	Deepak & Sivaswamy (2012) [37]	HEIMED,MESS DB	-	Motion Pattern, Rotational asymmetry metric	-	AUC = 0.99	-No Pre-processing and Post-processing steps
10	Deepak et al. (2012) [38]	MESS (522)	Rotation, Translation	GMP, LBP RT	-	DME, AUC = 0.99 SE = 100	-No segmentation at lesion-level & depend on normal images for training.
11	Ramasubramanian & Mahendran(2012) [39]	STARE DV(100)	RGB-HSI, MF, CLAHE	K-Means, GLCM, SVM	9	Success rate = 96%	-K-means clustering requires less time to process than Fuzzy C Means
12	Tariq et al. (2013) [42]	MESS(1200) STARE(81)	BE, CLAHE, NR	GF, Otsu thresholding, GMM	8	STARE,ACC = 97.9% SP = 98.32%	-Manual selection of threshold, Removal of OD eliminates false regions.
13	Adarsh Punnoili (2013) [40]	DBO(130) DV (40),STARE (100)	CN, GCE, CLAHE	MP, SVM, Triangulation method for macula detection	6	SE = 96.88% SP = 97.15%	-Accurate, distinct method for locating fovea.
14	Sreejinik & Govindan (2013) [41]	Private (90)	GCE, CN, MF	PSO, Otsu thresholding, MM	-	SP = 100% ACC = 93%	-By modifying the PSO parameters, exudate segmentation can be improved.
15	Chowriappa et al. (2013) [43]	MESS (100)	CE, MF, Histogram	Ensemble Model, NB, J48, SMO	14	ACC = 96.7%	-The system has been tested to accommodate a huge dataset.
16	Baby & Chandy (2013) [23]	MESS (1200)	GCE,CLAHE, ROI cropping	DT-CWT, GGD	104	Prec Rate = 78.23%	-The DT-CWT is appropriate for enabling CBIR.
17	Naguib et al. (2013) [22]	MESS (100)	GCE, BVR-Dilation	Edge Histogram Descriptor, CBIR	5	Prec Rate = 79.2%	-Low picture retrieval rate for DME
18	Chua et al. (2013) [27]	Private (300)	-	Texture features, FS	-	SE = 100% SP = 100%	-Satisfying results with fuzzy method(ACC = 86.67%)
19	Medhi & Dandapat (2014) [44]	DV(140) HRF(45) DB1(89)	CLAHE,BVR RGB-HSI	MM, Thresholding based on histograms	-	SE = 97.5% SP = 98.7%	-Unable to identify macula in non-macula-centric, poor quality images.
20	Vasanthi & Banu (2014) [28]	Private	GCE, MF, MIP	Adaptive Neuro-Fuzzy Inference, GLCM, ELM	5	SE, Fuzzy-100%, ELM-94.28%	-At high learning speeds, ELM appears to provide the best generalization.
21	Sharma et al. (2014) [31]	MESS DB(110)	CE, BVR	Histogram Thresholding, Positional properties	-	SE = 96.2% SP = 100%	-Pre-processing low contrast, non-macula-centered pictures may enhance results.
22	Ramya&Vijayprasad (2014) [46]	MESS (200)	GCE, CN, NR	Gaussian DD, PCA DD, Motion Patterns, RT	-	GaussianDD ACC-84% PCA DD-92%	-The PCA DD classifier yields better results than Gaussian DD.
23	Mookiah et al. (2015) [48]	MESS(300) Private(300)	CLAHE, Thresholding	Higher order spectra cumulants, RT Bispectrum magnitude	136	ACC,Private-95.9% MESS-95%	-Fine differences in the fundus picture pixels can be identified using the method.
24	Rahim et al. (2015) [24]	Private (600)	GC, Fuzzy HE	KNN, DT, Fuzzy Techniques	3	SE = 85.9% SP = 55.7%	-Fuzzy analysis facilitates improved picture representations.
25	Ibrahim et al. (2015) [26]	Private (300)	RGB-HSI, GC, MF	Discrete wavelet transform, Fourier spectrum, Fuzzy Classifier	25	ACC = 98.55%	-Easy interpretation of results using fuzzy rule-based system.
26	Kunwar et al. (2015) [47]	MESS (1200)	NR, AHE, CE	GLCM, PCA, SVM	8	SE = 91% ACC = 80%	-Texture features reduce the computation complexity.
27	Marin et al. (2016) [53]	MESS (1058)	-	Generation of ground-truth,Inter-observer Agreement	-	SE = 90.39% SP = 76.96%	-System functions with a SE equivalent to doctors.
28	Magotra et al. (2016) [74]	MESS (116)	GCE, NR, CLAHE	Spatial domain features Spectral features, SVM	10	SE values Severe cases-93%	-Features retrieved from only sub-regions than entire image.
29	Rahim et al. (2016) [25]	Private (600)	Fuzzy HE, filtering, GCE	RT, KNN, NB, Radial Basis Function(RBF) SVM	6	RBF&SVM,ACC = 93% SP = 93.62%	-All exudates were not segmented as a high threshold value is used.
30	Medhi&Dandapat (2016) [49]	DB,HRF DV, MESS	GCE, CE, CN	Top Hat Transformation,Otsu Threshold, Canny Edge Detector	-	SE = 98.86% SP = 98.05%	-The approach does not require the computationally expensive blood vessel removal.

Table 3 – Continued

SN	Author & Year	Datasets & Images	Preprocessing Techniques	ML Techniques	Number of features	Results	Observations
31	Medhi & Dandapat (2016) [50]	DB, HRF,DV(81) MESS(400)	CN, NR, Re	Spatial fuzzy clustering, Level set algorithm, Nick's threshold	-	SE = 98.6% SP = 98.9%	-Model could be used to estimate the boundaries of overlapping lesions,easy identification of dark and bright lesions
32	Mahaboob Shaik (2016) [51]	Private(50)	GCE, CE	Morphological image processing, Feed-forward neural network	4	ACC,SE more than 96%, SP = 100%	-Feed-forward NN outperforms probabilistic NN and all lesions were not considered.
33	Johny et al. (2016) [52]	Private(30)	BE, CE, GC, NR, HE	Localization of OD, fovea, Motion pattern generation	-	SE = 100% SP = 74%	-Superior in terms of computation time and ACC
34	Murugeswari&Sukanesh (2017) [45]	MESS	NR, CLAHE	Image thresholding, SVM, Cascade NN, Partial Least Square	8	ACC Values, SVM-98.33%, NN-97.16%	-Makes use of both fundus as well as OCT images for maculopathy detection
35	Lim et al. (2017) [75]	MESS (100)	CE, GCE	Completed LBP, Modified Completed LBP, SVM	-	SE = 80% SP = 70%	-Model is independent of the distance between lesions.
36	Acharya et al. (2017) [76]	MESS(300) Private(300)	CLAHE, AHE	RT, DWT, Discrete Cosine Transform, SVM	30	ACC,Private-100% MESS-97%	-No segmentation, top-hat, DWT help record subtle differences in pixel values.
37	Rekhi et al. (2017) [77]	MESS(100) DB1 (89)	CN, NR, GC, AHE	Morphological features, Anisotropic Diffusion, SVM	10	Highest ACC, DB-92.1%	-High ACC with a small false-positive rate,suitable for real-time screening.
38	Punniyamoorthy et al. (2018) [78]	DB1 (89)	NR, GCE, CE	Morphological operations, Image processing techniques	-	SE = 97.75% ACC = 98.86%	-Obtained encouraging results with image processing techniques
39	Marin et al. (2018) [79]	MESS (1058)	SC, NR, Thresholding	Generation of ground-truth, Regularised Local Regression	9	SE = 90% SP = 69.39%	-High SE for cases needing urgent intervention.
40	Zhou et al. (2018) [80]	MESS (1200)	Re, MM, BE	Fisher Discriminant Analysis, Superpixel segmentation	-	SE = 96.46% SP = 98.56%	-Improved red lesion detection performance.
41	Syed et al. (2018) [81]	MESS, DV Private(462) DB1(89)	Thresholding, MM, HE, NR	Hybrid features, Gabor wavelets Multilayer thresholding, SVM, Minimum distance classifier	4	ACC, DV-100% DB1-94.3% MESS-93.50%	-For appropriate fovea localization, a new set of features is combined with a minimum distance classifier
42	Ren et al. (2018) [82]	MESS(500) E-ophtha(83)	-	Macula localisation, Anatomical features, Graph based classifier	23	ACC = 97.5% F1-score = 0.942%	-Macula localization along with exudate identification through vector quantization.
43	Sri et al. (2019) [83]	MESS (65)	Inversion, Scaling	Automatic Distance Calculation, Area estimation of exudates	6	ACC = 78.47%	-ACC can be enhanced with the use of a larger dataset
44	Thulkar&Daruwala (2019) [84]	DB1 (89)	Re, GCE, HE, CLAHE	ANN, Morphological filtering, Thresholding	6	ACC = 97.18%	-ANNs have been established to be useful tools for classifying features.

Legend: MESS = MESSIDOR, DB = DIARETDB, DV = DRIVE, BE = Background Estimation, GCE = Green Channel Extraction, HE = Histogram Equalization, AHE = Adaptive Histogram Equalization, ACC = Accuracy, SE = Sensitivity, SP = Specificity, Prec = Precision, AUC = Area Under the Curve, ROI = Region of Interest, CLAHE = Contrast Limited Adaptive Histogram Equalization, IE = Intensity Enhancement, MIP = Morphological Image Processing, CE = Contrast Enhancement, NR = Noise Removal, Re = Resize, Au = Augmentation, GC = Grayscale Conversion, BVR = Blood Vessel Removal, CN = Color Normalization, IR = Image Rotation, IC = Illumination Correction, LBP = Local Binary Patterns, SVM = Support Vector Machine, PCA = Principal Component Analysis, HT = Hough Transform, GF = Gaussian Filter, GaF = Gabor Filter, CBIR = Content Based Image Retrieval, MLP = Multi-Layer Perceptron, KNN = K-Nearest neighbour, NB = Naive Bayes, RF = Random Forest, GMP = Generalized Moment Patterns, RT = Radon Transform, MF = Median Filtering, RGB-HSI = RGB to HSI Conversion, GLCM = Gray Level Co-occurrence Matrix, GMM = Gaussian Mixture Model, PSO = Particle Swarm Optimization, SMO = Sequential Minimal Optimization, DT-CWT = Dual-Tree Complex Wavelet Transform, GGD = Generalized Gaussian Density, FS = Fuzzy-Sugeno, OD = Optic Disc, MM = Mathematical Morphology, ELM = Extreme Learning Machine, DD = Depth versus Depth, PCA = Principal Component Analysis, DT = Decision Tree, DWT = Discrete Wavelet Transform, SC = Shade Correction.

related to clinical trials carried out to understand the disease progression in the medical domain, were excluded. Only articles focusing on computerized DME detection were included, with all of those addressing other eye disorders besides DME being excluded. To offer a thorough assessment of the state-of-art computerized DME detection systems, the research examines the literature from the perspectives stated below.

1. CAD techniques for DME detection employing CFP.
2. CAD techniques for DME detection employing OCT.

Some of the authors used traditional ML techniques on fundus and OCT images for the detection of DME, whereas others used DL methods. A detailed review of various methodologies used by different authors in the classification of DME in the retinal fundus and OCT images is presented. To integrate the relevant publications in a more comprehensible manner, this review employs a meticulously designed structure.

2.1. CAD techniques for DME detection employing CFP

DME in fundus images is characterized by the presence of lesions such as hard and soft exudates (EX). Assessing the position of the macula, fovea, OD, or lesions near the macula is critical for rapid recognition and investigation of the DME. Many investigations have been performed to determine DME from CFP using various ML-based algorithms, which are discussed in this section.

2.1.1. Traditional ML-based methods

In the field of medical image processing and computer vision, ML plays a critical role. The steps involved in an image processing pipeline include pre-processing, segmentation, feature extraction, and classification. The following sections throw light on the various ML-based models listed in Table 3.

2.1.1.1. Methods based on Content Based Image Retrieval (CBIR). Rather than learning the lesion patterns (HEs, hemorrhages, and microaneurysms), the anomalous regions are automatically detected by learning the normal background and the content in the case of CBIR systems. Diseased regions are identified as those that differ significantly from the regions in the normal images [21]. Naguib et al. [22] proposed an automated CBIR model that extracts fundus images with DME determined by the number and region of exudates by partitioning the macula into three concentric zones. Next, the textural inconsistencies of these areas were used to indicate the lesion content of the retina. Since, low-level features like color, texture, and shape were used by CBIR systems, they were unable to investigate and interpret the visual context of the image semantically with low image retrieval rates. Further, wavelet-based image features were used because of their shift-invariant and directionally selective attributes, for faster image retrieval [23].

2.1.1.2. Methods based on fuzzy techniques. Fuzzy rule-based classifiers have spurred a great deal of interest in image classification with decision support systems because of their simplicity of interpretation. Rahim et al. [24] employed pre-processing of the fundus images using fuzzy techniques to extract three features, namely, the mean, the area and the

standard deviation of pixels intensities followed by classification using a few ML methods. They also used a blend of fuzzy approaches, including fuzzy filtering, fuzzy histogram equalization, and finally fuzzy edge detection along with three different techniques for macula localization [25]. Research in later years showed the interpretability of fuzzy rule-based systems, when combined with the accuracy of neural networks provides good classification performance [26]. In [27], texture features were extracted and those features which were significant statistically, were used for fuzzy classification. Further evolution of trends in the literature proved that the extraction of features using ML-based methods such as texture analysis, and segmentation followed by the use of fuzzy classifiers could help obtain very good results [28].

2.1.1.3. Morphology, texture and intensity feature (MTIF)-based methods. Every pixel in an image was updated depending on the value of other pixels within the proximity in a morphological operation. The majority of the authors have employed morphological image processing to locate or segment macula, fovea, OD, and exudates. The information obtained was used to grade DME employing various ML classifiers. Nayak et al. [29] aimed at the detection of OD, fovea and exudates. The OD was identified using the variation in the intensity of adjacent pixels. Statistical measures such as standard deviation and correlation were used to identify the fovea. Areas of exudates were extracted from the fovea, foveola, perifovea, and parafoveal regions as four characteristics. OD and cup are the brightest of all regions in a retinal fundus image. In [30,31], the fundus image was iteratively thresholded to identify OD based on the number of pixels that are needed to form OD. The macula's position and distance from the OD remain relatively unchanged. Macula was also located based on a distance of 2-disc diameter temporal to OD. Exudates were precisely segmented using dilation and their position information was used to grade DME.

Combining a strategy based on markers to control over-segmentation with an exudate detection algorithm, Lim et al. [32] were able to locate the fovea and grade DME severity. In [33], the area of exudates in the macular region is computed to help grade classes such as clinically significant macular edema (CSME), non-CSME. Hunter et al. [34] calculated the maximum contrast and peak points on exudate lesions, to screen referable maculopathy. For the identification of OD, macula, and exudates, Akram et al. [35] used morphological methods including opening, closing, dilation, and erosion. Exudate and fovea detection was performed utilising Kirsch edge operator as well as the region-growing method by Giancardo et al. [36]. They graded DME employing wavelet decomposition and the probability map for exudates. A feature extraction method has been used to capture the global features of the fundus images and distinguish normal from DME images by Deepak and Sivaswamy [37]. Deepak et al. [38] employed approaches inspired by the impact of motion on the perception of structures, which eliminated the need for the segmentation of lesions.

In [39], the regions in a fundus picture were subsequently split into five clusters with K-means using the Euclidean distance measure. Colour and texture features were extracted from the segmented clusters for classifying them into exu-

dates and non-exudates. Lesion-level segmentation is tedious and time-consuming as it requires image annotations from expert ophthalmologists. In [40], the primary retinal vascular arcades' position and the geometrical difference between the OD and fovea were used to locate the fovea. In the second stage, exudates and haemorrhages were detected using morphological operations. In [41], the OD was determined using mathematical morphology, and the fovea was detected using an annular mask in addition to the segmenting of the exudates with the Particle Swarm Optimization approach.

The technique by Tariq et al. [42], utilizes the vascular anatomy, location of OD and filter bank to extract the macula and potential exudate areas. The Gaussian Mixture Model was then used to grade maculopathy using macular coordinates and feature vectors for each exudate region. Further research presented a decision-making approach that uses an ensemble of four prominent classifiers to enhance the performance [43]. Few approaches aimed at the elimination of blood vessels to enhance the detection of macula [44,45]. OD and macular regions were both located by the illumination variation within pixels along with the use of bounding boxes created for the specific region by Ramya et al. [46]. They converted the fundus image to a motion pattern, which is an intermediate representation that spatially locates exudate existence independent of their size. As there is a noticeable change in the texture in the region immediately surrounding the macula, one disc diameter apart, Kunwer et al. [47], employed the extraction of eight texture features for DME grading. Higher-order features including the third-order spectrum and moments of radon projections computed on grey-scale lesions were extracted by Mookiah et al. [48]. Various supervised classifiers were used to grade DME based on the extracted features.

Medhi and Dandapath [49] processed the information about the relation of blood vessels with OD and macula and various colour planes to identify lesions such as microaneurysms, haemorrhages and exudates. In another approach, these authors detected fovea (with accuracy = 98.84%) by cropping the macular neighbourhood [50]. The method by Shaik et al. [51] isolates the macula from the retinal fundus image using the location of the OD and the vascular anatomy to first segment the iris. Using filter banks, a binary map for potential exudate regions was generated to provide a comprehensive feature vector per each region. Employing the creation of motion patterns, Johnny et al. [52] reported a computer-based method for DME identification in which, the macula was initially located utilising the typical distance from the OD. Binary smear patterns were then produced using the location of the OD and motion patterns. The system's effectiveness was evaluated using various datasets, and the MESSIDOR dataset achieved the greatest AUC of 0.99. Marin et al. [53], segmented OD based on the variation in the intensity values of the pixels. They were able to detect tiny exudates by focusing on a circular region of interest and an exudate probability map that helped them to grade DME.

Subsequent research in the domain presents hybrid feature extraction approaches used to improve the accuracy of classification [76]. In order to process the specific sub-regions of interest rather than processing the entire image, Magotra et al. [74] evaluated features in both the wavelet

and spatial domains. The method by Rekhi et al. [77], identifies macular regions employing morphological operations, and segments exudates using SVM. The location data of exudates in these macular regions was used to grade the severity of DME. To acquire encouraging results for DME classification, Lim et al. [75], used textural feature analysis with a modified CLBP technique to extract local gray-level information of all channels. Using 89 fundus images from the DIARETDB1 dataset, Punniyamoorthy et al. [78], Zhou et al. [80], employed a number of image processing approaches, such as contrast enhancement, followed by masking and edge-detection to detect the OD, exudates, and the presence of macular edema.

For appropriate foveal localization, a completely new set of features is combined with a minimum distance classifier by Syed et al. [81]. They graded DME based on the distance of exudates from the macula. To identify areas that are anticipated to have exudates, Marin et al. [79], applied digital image processing techniques to the fundus pictures. To detect the regions thought to be real exudates, supervised classifiers were used. Ren et al. [82] used a vector quantization technique for segmentation of exudate lesions and identification of macula. Sri et al. [83] considered the characteristics such as the distance between the macula and exudates, the area of exudates, static mean and standard deviation to detect DME. Further improvements in the field were achieved by the use of morphologically extracted features along with Artificial Neural Networks (ANNs) to obtain better classification accuracy [84].

2.1.2. DL-based approaches

DL is a sub-field of ML that employs multi-layer neural networks to learn from massive amounts of data. The state of art papers that have used various DL models for processing retinal CFP is depicted in Table 4.

2.1.2.1. Methods based on custom Convolutional Neural Networks (CNN). Convolutional, pooling, and fully-connected layers are indeed the major types of layers that constitute CNNs. The power of CNN rests within the deep architecture, capable of retrieving unique features automatically at multiple levels of abstraction [85].

AlBander et al. [86] trained three CNN architectures with three different image sizes of 128×128 , 256×256 and 512×512 pixels respectively to classify DME with less computation time. The layer weights of the subsequent networks were initialized from the layer weights of the previous networks. A framework by Auccahuasi et al. [87], enabled automatic exudate recognition in retinal fundus images using CNN and a dictionary of images with and without hard exudates. AlTurk et al. [88], employed CNN-based automated software for localisation of exudates for computerized DME categorization. To tackle the difficulties of significant intra-class variability and strong interclass similarity a convolutional residual network is employed to diagnose DME by Mo et al. [89]. Li et al. [90] developed four DL algorithms each based on Inception v3 architecture for the identification of referable DR, DME, and an assessment of image quality for DR and DME. To diagnose DME at the image, eye, and patient levels, a binary classification neural network was trained by Bressler et al. [91], with hyperparameter tuning across the val-

Table 4 – Use of fundus images with DL.

SN	Author & Year	Datasets/Images	Techniques Used	Results	Observations
1	Gulshan et al. (2016) [54]	EyePACS(9963) MESS2(1748)	Deep CNN, Inception-v3, TL, SGD	EyePACS SE = 90.3% SP = 98.1%	-Model may underperform for images that most of the doctors would miss.
2	Perdomo et al. (2016) [100]	E-ophtha(47) MESS	AlexNet, Exudate-detection	ACC = 77% with a batch size of 32	-Efficient in finding NORMAL class.
3	AlBander et al. (2016) [86]	MESS (1200)	Feature learning, Deep CNN Network, design and training	SP = 96.5% ACC = 88.8%	- No burden of segmentation and handcrafting features.
4	Mo et al. (2018) [89]	E-ophtha(82) HEI-MED(169)	Cascaded deep residual networks, Exudate-segmentation	AUC, HEI-MED-0.9709 Eophtha-0.9647	-Avoids pre and postprocessing to minimize the influence of subjective factors.
5	Li et al. (2018) [90]	Private (1,06,244)	CNN, The consensus grading, Internal Hold-Out Validation	External validation, AUC = 0.95, SP = 98.5%	-Method has the power to improve DR screening productivity.
6	Sahlsten et al. (2019) [101]	Private (41122)	TL method, Inception-v3	AUC = 0.989	-DL potentially improves the cost-effectiveness of diagnosis.
7	Arcadu et al. (2019) [107]	Private (17,997)	DL regression model, Inception-V3, TL cascade	For CST, AUC = 0.97 For CFT, AUC = 0.91	-Generalization of model is difficult with clinical trial dataset.
8	Harangi et al. (2019) [95]	EyePACS(22700) IDRiD(516)	Ensemble of CNN, Traditional hand-crafted features, AlexNet	3-class DME ACC = 96.85%	-Quantitative comparison of models, Extracted 68 features.
9	He et al. (2019) [108]	MESS (1200) IDRiD(516)	U-net model, VGG-16 Multi-scale Feature Extraction	ACC = 96.3%, AUC = 0.98 SP = 97.12%	-Good results for MESS.
10	Raumviboonsuk et al. (2019) [96]	Private (25,326)	CNN, Inception-v4, Ensemble model, Cascade of thresholds	SE = 96.80%, AUC = 0.987 SP = 95.60%	-Pictures that are ungradable by the graders are excluded.
11	Al Turk et al. (2020) [88]	EyePACS(28100) DB1 Private(56658) MESS2(1748)	CNN framework Analysis of disease Progression, External and internal, Validation, Lesion detection	Agreement between the automated software and human grading for all datasets was between 0.84 to 0.88	-The software can assist image processing in diabetic patients across a wide range of demographics.
12	Auccahuasi et al. (2020) [87]	DB1(89) Private(906)	Deep CNN, Creation of a dictionary of images, DME-detection model	SE = 92% SP = 93%	-A new means of detecting HEs in retinal photographs automatically.
13	Chalakkal et al. (2020) [102]	MESS(1187) IDRiD, Private(200)	Pre-trained CNN, Meta-heuristic feature selection, KNN based classifier	Model's ability to deal with class imbalance is discussed	-Effects of skewed data, can be resolved by feature space over-sampling.
14	Sulaiman et al. (2020) [112]	IDRiD(516)	Convolutional autoencoder network, CLAHE, Dataset augmentation	ACC = 68% Prec = 66%	-Class weights and augmentation approaches can help avoid class imbalance.
15	Varadarajan et al. (2020) [109]	EyePACS(990) Private(7072)	Deep CNN, Inception-v3	ROC-AUC = 0.89, SE = 85% SP = 80%	-OCT-based DME attributes predicted more accurately than doctors.
16	Singh et al. (2020) [85]	IDRiD(516) MESS	DMENet, Hierarchical Ensemble of CNNs	ACC = 96.12% SE = 96.32%	-Good results with sophisticated preprocessing step.
17	Sarki et al. (2020) [103]	HEI-MED (169)	Pre-trained CNN, Diabetic eye disease detection	Issues in public datasets discussed.	-Model determines all types of DEDs, those in the early stages.
18	Yu et al. (2020) [92]	Private(3496)	Automated software, CNN Retrospective analysis	Conbercept treatment outcome analysis	-A comprehensive, unbiased method for tracking DR.
19	Wu et al. (2020) [94]	EyePACS, HEIMED, MESS, E-ophtha(3496)	Positional relationships, DL MD-ResNet, Multi-level feature fusion	E-ophtha, ACC = 87% HEI-MED, ACC = 94.4%	-The enhanced feature information aids the exudate segmentation to be accurate.
20	Li et al. (2020) [113]	MESS (1200) IDRiD(516)	Cross-disease Attention Network	DME, AUC = 0.924 ACC = 91.2%	-On the MESS dataset, it outperforms other techniques.
21	Nazir et al. (2020) [93]	MESS, DIARET DB1(89), HRF(45) DRHAGIS(40)	FRCNN, Fuzzy K-Means, Bounding-box annotations, Multi-Task Loss function	DME, SE = 96% SP = 95.8% ACC = 95.8%	-Integrated approach for diagnosing glaucoma, DME, and DR which has disease-specific symptoms
22	Ahn et al. (2021) [98]	IDRiD(516) Private(44 55) EyePACS(30000)	Lesion, Vessel segmentation mask, Conditional GAN	DR, F1-score = 0.74 DME, F1-score = 0.91	-The model could not generate information from fundus images with uniformity.
23	Kang et al. (2021) [110]	Private (5117)	Deep CNN, EfficientNetB4, Multimodal imaging	DME AUC = 0.995 SP = 98.8%	-Performed well in detecting diseases requiring early treatment.
24	Li et al. (2021) [97]	Private(8739) MESS2	Multi-Inception-v4 ensembling, SGD	Referable DME, AUC = 0.99 SP = 97.1%	-Analysis of effect of number and size of training images on result.
25	Reddy et al. (2022) [114]	IDRiD(516)	ResNet50, Disease-specific, Channel and spatial-wise attention features	ACC Values, DR = 96%, DME = 93.2%, DR-DME = 92.2%	- Optimal feature selection with reduced loss functions and attention method.
26	Wang et al. (2022) [111]	Private(35001) MESS(1200) MESS2(1748)	EfficientDet-d1, Bi-directional Feature Pyramid Network	AUC values, Private = 0.981 MESS = 0.952 MESS2 = 0.958	- Due to the presence of other disorders such as OD edema, precise visualization of OD was difficult.
27	Kamble et al. (2022) [99]	MESS(1200) IDRiD(516)	Lesion-aware network, GAN-based method, U-Net based segmentation	AUC values, IDRiD = 0.925 MESS = 0.942	-Individual lesion-based approaches were outperformed by the model.
28	Liu et al. (2022) [104]	Private(1,167,791) CFP & OCT images	Deep CNN trained on Tensorflow, EfficientNet-B5	SP = 80%, SE = 81%	-DLS could successfully generalize to a wide range of diverse ethnicities.
29	Bressler et al. (2022) [91]	EyePACS(32,049)	Binary classification Neural Network, Hyperparameter Tuning	At Patient level- SP = 90%, SE = 90.1%, AUC = 0.962	-The detection of DME was done at the patient, imaging, and ocular levels.
30	Kumar et al. (2022) [105]	IDRiD (680)	pre-trained DenseNet121, Adam optimizer	ACC = 86.4%	-Model performed better than the state-of-the-art models presented during the ISBI-2018 challenge.
31	Kumar et al. (2022) [106]	IDRiD	RadioDense model, DenseNet121, Gradient Boosting	ACC = 87.4%	-The feature extraction results with gradient boosting surpass other classifiers.

Legend: MESS = MESSIDOR, DB = DIARETDB, CNN = Convolutional Neural Network, TL = Transfer Learning, SGD = Stochastic Gradient Descent, DL = Deep Learning, VGG = Visual Geometry Group, KNN = K-Nearest Neighbour, CLAHE = Contrast Limited Adaptive Histogram Equalization, FRCNN = Fast Region-based Convolutional Neural Network, GAN = Generative Adversarial Network, ACC = Accuracy, SE = Sensitivity, SP = Specificity, Prec = Precision, AUC = Area Under the Curve, DME = Diabetic Macular Edema, DR = Diabetic Retinopathy, ROC- Receiver Operating Characteristics, OCT = Optical Coherence Tomography, CST = Central Subfield Thickness, CFT = Central Foveal Thickness, HE = Hard Exudate, DED = Diabetic Eye Disease, OD = Optic Disc, DLS = Deep Learning System.

idation set. Yu et al. [92] validated DL models to automatically recognize and measure intraretinal hemorrhage, and HEs in CFPs of DME patients receiving intravitreal conbercept therapy.

2.1.2.2. Methods based on object detection. Localized diseased regions were segmented using the bounding box notation with Fast Region-based CNN (FRCNN), rather than segmenting the entire image to reduce computation. Nazir et al. [93] presented a method based on FRCNN that locates the bounding boxes of abnormal, diseased areas in the fundus image. Results prove that FRCNN and Fuzzy K-means work together to precisely locate the areas that were affected, ensuring accurate detection of DME. Wu et al. [94] combined dilated convolution with a multi-level feature fusion technique for accurate and reliable segmentation of hard exudates using MD-ResNet, a DL-based residual network.

2.1.2.3. Ensemble of DL classifiers. The ensemble framework combines a variety of learning strategies to produce better prediction outcomes than that a single learning strategy. Many approaches incorporated the ensemble of several CNNs for the classification of DME. In an article by Harangi et al. [95], an ensemble approach for eye image classification which incorporates a CNN with traditional hand-crafted features into a single architecture is used. The purpose of the study by Raumviboonsuk et al. [96] was to evaluate a DL algorithm on a massive patient group and compare its performance to that of professional graders. Singh et al. [85] evaluated fundus pictures with the help of an ensemble of CNNs along with ways to address class imbalance issues in the dataset. Li et al. [97] developed an ensemble of five models with each model being built based on improved Inception-v4 architecture for classification of DR and DME.

2.1.2.4. Methods based on Generative Adversarial Network (GAN). GANs are a DL-based approach to generative modeling. Ahn et al. [98] synthesized future retinal CFPs to obtain vessel and lesion data by using Conditional GANs. The authors visually illustrated the ability of the fundus image generator to predict the progress of the disease within the fundus images generated by the model. Kamble et al. [99] developed an adversarial, lesion-aware model for DME classification that integrates feature representation and learning.

2.1.2.5. Transfer-Learning (TL) methods. TL seeks to preserve and apply the information gathered when tackling one issue to a different but related one. TL can be adopted when there is limited training data available for medical image classification. Gulshan et al. [54] detected DR and DME by using an Inception-v3 architecture with batch normalization and pre-initialization of weights to accelerate the training. The method was tested at two different operating settings, one each for high specificity and high sensitivity. Perdomo et al. [100] offered AlexNet based architecture for localisation of exudates and DME categorization as an assistance for DR diagnosis. Sahlsten et al. [101] developed a DL algorithm based on artificial intelligence for the identification of referable DR. Chalakkal et al. [102] attempted to address two primary challenges: class imbalance concerns and segment-

ing out exudates by integrating a pre-trained deep CNN with meta-heuristic feature engineering.

Sarki et al. [103] attempted the early detection of glaucoma, DME, and DR using a single approach. In [104], authors generalised a DL system that used CFP as input to predict ground truth derived from OCT scans across various screening cohorts. They also looked at the capacity of the system to identify centre-involved DME using definitions based on retinal thickness and IRF occurrence. In [105], CFPs were utilised to train a DenseNet121 to extract the useful set of feature vectors, which were then passed into a few further fully connected layers and then onto the classification layer to categorise DME instances. Further research demonstrates an efficient implementation of cutting-edge technology, such as the RadioDense model for identifying DME from CFPs [106]. Arcadu et al. [107] used a TL cascade to anticipate time-domain OCT (TD-OCT) equivalent metrics of macular thickness, such as central subfield thickness (CST) and central foveal thickness (CFT). He et al. [108] employed TL along with XGBoost and SVM for DME classification. Varadarajan et al. [109] used TL for predicting center involved-DME along with Intra Retinal Fluid (IRF) and Sub Retinal Fluid (SRF). A DL model was developed by Kang et al. [110] incorporating multimodal imaging techniques. Wang et al. [111] evaluated the performance of an end-to-end TL fusion model for DME classification and HE visualization on an edge device.

2.1.2.6. Auto Encoder (AE) based methods. Autoencoders condense the input into a latent-space description, that is subsequently used to reconstruct the output. Sulaiman et al. [112] used an autoencoder network along with class weights and image augmentation to address concerns of class imbalance.

2.1.2.7. CNN models with Attention Mechanism (AM). The interdependence between input and output variables, and within the input variables, is managed and quantified by attention, which is one component of a network's architecture. By investigating the internal relationships between the diseases with only picture-level supervision, Li et al. [113] collectively graded DME and DR with attention modules. Reddy et al. [114] used two attention modules to extract disease-specific features for DR and DME which aided in the joint DR-DME classification.

2.2. CAD techniques for DME detection employing OCT images

Doctors frequently employ retinal OCT scans to diagnose various eye disorders such as DME, Macular Hole (MH), Age-Related Macular Degeneration (AMD), Choroidal Neovascularization (CNV), and so on. CAD solutions that process OCT images use various ML and DL-based algorithms to significantly speed up disease detection. The following section elaborates on various CAD systems used in the detection of DME by further categorizing the models based on classical ML, DL approaches along with their subcategories. Tables 5 and 6 illustrate various OCT-based, ML and DL methods used for DME detection.

Table 5 – Use of OCT images with traditional ML.

SN	Author & Year	Datasets & OCT Volumes	Preprocessing Techniques	ML Techniques	Number of features	Results	Observations
1	Liu et al. (2011) [115]	Private(457 B scans)	Th, MF, MIP	MSSP, LBP histogram, SVM Labeling by three doctors	-	Overall AUC $\geq$ 0.94 ME, AUC = 0.985	-Instead of analyzing an entire OCT volume, single foveal slice is used.
2	Wilkins et al. (2012) [116]	Private 16 patients	GC, MF, SNR-balancing	Bilateral Filtering, Th, Boundary Tracing	-	For CME, SE = 91% SP = 96%	-The segmentation algorithm's ACC is compromised by low SNR.
3	Roychowdhury et al. (2013) [128]	Private (120 images)	Scaling, MF, ROI extraction	Segmentation of 6 sub-retinal layers, Iterative high-pass filtering, Cyst identification	3	ACC Values: Inner plexiform-88%, Inner nuclear-86%	-Cyst detection result ($= 0.9$) is highly correlated with manual detection result.
4	Srinivasan et al. (2014) [66]	-Public Dukes(45)	BM3D, RCF, IC	Multiscale HOG, SVM	-	For DME, AMD ACC = 100%	-Entire volume examined, no layer segmentation complexity
5	Lemaître et al. (2015) [117]	Private(3800) -Dukes (45)	NLM Filtering	LBP, DiL using BoW, RF Global, local mapping	6	SER, SE = 87.5% Dukes, SP = 100%	-Results prove that features from LBPs exhibit strong discriminative ability.
6	Sugruk et al. (2015) [129]	Private (16 images)	NR, GF	Drusen and bubble detection	-	ACC = 87.5%	-Small dataset used which is insufficient for clinical validation.
7	Hernandez-M et al. (2015) [133]	Private (277 eyes)	-	Binary LR, Non-linear equations, Intraretinal cyst detection	2	AUC = 0.88, SE = 83%, SP = 89%	-Detection of retinal cysts resulted in accurate detection of DME.
8	Chiu et al. (2015) [71]	Dukes2 6 patients	Re, Normalize, KR based NR	KR, GT, DP	12	Overall mean Dice coefficient = 0.78	-A high quality dataset can be used to enhance results
9	Syed et al. (2016) [130]	Private (90)	-	Reconstruction of 3D retinal thickness surfaces, SVM	8	ACC = 98.88% SE = 100%	3D macular thickness can be used to verify cyst volume.
10	Wang et al. (2016) [119]	Dukes(45)	-	LCP, SMO, LR, NB, MSSP	23	DME&NORMAL, ACC = 100%	-Building small feature subsets need less time and help avoid over-fitting.
11	Lemaître et al. (2016) [118]	Private (3800 B scans)	Flatten, NLM, Aligning, MF	LBPs, BoW, RF, SVM, KNN, LR, GB	-	RF, SE = 81.2% SVM, SP = 93.7%	-With various classifiers, feature setups, results of preprocessing appear unclear.
12	Hassan et al. (2016) [131]	(Private) (90 scans)	Re, GC, CE, NR, MIP	Coherent 2D tensor, 5D feature vector, Thickness Profile, SVM	5	ACC = 97.77%, SE = 100%	-Method is rotation invariant, does not rely on the curvature of the retina.
13	Girish et al. (2016) [136]	-Public Optima(30)	Bayesian NLM	Marker controlled watershed transform, K-means clustering	-	SP = 93.33% Recall rate = 82% Prec rate = 77%	-Less post-processing relative to conventional watershed.
14	Arif et al. (2017) [121]	Private (75 images)	GCE, MIP, Re	Cyst Segmentation, Discriminant Analysis, Otsu algorithm	4	DME, ACC = 100%	-The method needs to be validated with a larger dataset.
15	Alsaifi et al. (2017) [120]	Private (3800 B scans)	BM3D, RCF, IC	HOG and LBP, PCA and BoW Linear, Kernel SVM, RF	347,940	NORMAL, ACC = 91.8% SE = 87.5% SP = 87.5%	-Preprocessing results are unclear with different classifiers.
16	Sidibe et al. (2017) [137]	Private(32) Dukes(45)	NLM, NR, RCF	GMM, Anomaly detection, PCA	-	Private, SE = 93.7% Dukes, SP = 100%	-Model detects abnormal B scans within a OCT volume.
17	Sun et al. (2017) [122]	Dukes(45) Private (678)	Align, NR, IC BM3D, MF	Sparse coding, DiL, Feature Representation, SVM	-	Dukes, ACC = 100% Private, ACC = 99.6%	-Dictionary-learning phase of the model is not very effective.
18	Naz et al. (2017) [134]	Duke2(108) Dukes (45)	NR, BM3D	2D structure tensor MIP, SVM	4	Private ACC = 79.65%	-SVM outperformed Naive Bayes and KNN.
19	Fang et al. (2017) [138]	Dukes(45) [66] Private (54)	Re, BM3D, Flatten	PCA Network with Composite Kernel (PCANet-CK), ELM	-	ACC, SE, SP-100%-Dukes	-Block, PCA filter size is assessed for best results.
20	Fang et al. (2017) [234]	-Private(86)	RCF, BM3D, NR	SURF, BoW, SVM	64	Average ACC = 88.72%	-The method can detect multiple associated lesions.
21	Hussain et al. (2018) [124]	-Dukes(45)-Private	-	Retinal features, LR, SVM, NB, RF, DT	10	2-class, ACC = 96%	-Thickness of all layers is not used, it fails for images with multiple lesions.
22	Dash and Sigappi (2018) [125]	Private (150 B scans)	Cr, GC, NR, MF	Graph-cut search, DP, LBP, SVM, Cascade NN	8	AUC = 99% For Cascade NN SP = 100%, ACC = 96%	-Cascade neural networks outperformed SVM with F-Score = 97%
23	Dash and Sigappi (2018) [126]	Private (80 images)	Flatten, Cr, GC, NR, MF	SURF, GF, Hessian matrix, SVM	64	Average disease detection ACC = 99%	-SURF with SVM tends to provide good performance.
24	Girish et al. (2018) [139]	-Optima	NR, Unbiased bilateral filter	Modular way to CME detection-Exhaustive analysis	-	Comparative study of CME detection models	-Layer segmentation helps in cyst identification.
25	Rashno et al. (2018) [142]	Private(29)-Dukes-(10)-Optima(4)	Gamma-correction, NR	CME, SRD detection, Neutrosophic Domain, DP	-	SE, Optima-89% Dukes -67%, Private-76.7%,	-Abnormal intensity variations in layers may lead to unreliable results.
26	Samagaio et al. (2018) [141]	-Private (560 B scans)	-	Intensity, texture and clinical features, GT, DP, KNN, NB	307	For 7-KNN, JI=0.6625 DC-0.78, ACC = 93.6%	-7-kNN classifier outperformed other classifiers along with post-processing.
27	Moura et al. (2018) [20]	-Private (38 scans)	NR, Shadow Removal, CE	GT, DP, MIP	9	SE = 89.30% JI = 0.7916, DC = 0.83	Four visualization modules were created to help doctors easily detect SRD.
28	Samagaio et al. (2018) [150]	-Private (170 images)	NR, MF	GT, DP, SVM, NB, SRD & CME-MT, DRT-LS	27	F-measure, DRT-87% CME-91%, SRD-100%	-Detects DME even with 3 subtypes exist intertwined within the retinal layers.
29	Samagaio et al. (2018) [149]	-Private (50 images)	NR, MF	SRD & CME-MT, DRT-LS	27	F-measure, DRT-83%, CME-81%, SRD-100%	-Detects DME even with 3 subtypes exist intertwined within the retinal layers.
30	Athira et al. (2018) [123]	-Private (Images of 45 patients)	NR, MF	Texture Analysis, GT, Quadratic Discriminant Analyzer	7	ACC = 97.5%	-The approach can be enhanced with DL based methods.

Table 5 – Continued

SN	Author & Year	Datasets & OCT Volumes	Preprocessing Techniques	ML Techniques	Number of features	Results	Observations
31	Chen et al. (2018) [140]	-Private (15 images)	NR	Improved level set, Anisotropic diffusion	-	SE = 86.9%, Precision = 81.1%, DSC-0.80	-SNR of all images was balanced for consistent results.
32	Moura et al. (2019) [143]	-Private (70 images)	-	Delimitation of ROI, GLCM, HOG, GF, LBP, SVM, DRT-LS	307	Jl = 0.8346 DC = 0.9051	Post-processing the vascular areas help address misclassifications.
33	Otero et al. (2019) [151]	-Private (96 images)	-	GF, HOG, LBP, Linear Bayes Normal, KNN	50	ACC, SRD-93%,DRT-88.8%, CME-90%	-The model was tested on a very small dataset with no preprocessing.
34	Hassan et al. (2019) [132]	-Public BIOMISA(90)	Normalize,GCE, Wiener filter	Computation of coherent tensors, 7D feature vector, SVM	7	ACC = 97.78% SP = 100%	Authors graded DME into CSME and non-CSME
35	Moura et al. (2020) [146]	-Private (40 OCT scans)	-	Layer Segmentation, Identification of ROI, ML strategy	-	Good results, for foveal, perifoveal, parafoveal areas	The method can detect DME types in foveal, perifoveal and parafoveal regions.
36	Wu et al. (2020) [147]	-Private (10)	NR, Bilateral filter	GMM clustering, Spatial data, Improved level-set	-	For CME DC=74.89 ± 3.91	-Issues in layer segmentation may hinder the performance.
37	Moura et al. (2020) [144]	-Private (262 images)	-	CME,SRD-Graph cuts,Active contours, DRT-LS	354	DC values, DRT-0.89, CME-0.74, SRD-0.87	-For CME,SRD, preliminary identifications were not acquired precisely.
38	Mousavi et al. (2020) [127]	-Dukes(45) [66]	NR, IC, RCF	DiL, HOG, Sparse representation	-	ACC, DME-95.13% NORMAL-100%	-Fisher Discrimination DiL and HOG provide good results.
39	Moura et al. (2020) [145]	-Private (170 scans)	-	GLCM, SVM, NB, CME, SRD-Multi-level Th, DRT-LS	18	F-measure,SRD-100% CME-91.9%,DRT-84%	-Instead of handwritten features, DL models can be used to detect features.
40	Wang et al. (2020) [148]	-Private (100 images)	-	K-means clustering, Geodesic Active Contour	-	Prec = 97.7% SP = 99.2%	-Model needs to be improved for low contrast images.
41	Maurya et al. (2020) [135]	-Private (50 images)	Flatten,GC, NR,MF,BM3D	Gradient information, Retinal thickness data, Pixel intensity values	-	Measured retina thickness,set a threshold to detect edema	-Detection of all edema types of DME is not incorporated.
42	Girish et al. (2020) [152]	OPTIMA [68](8 OCT volumes)	NR, Bayesian NLM	Topographical watershed transform, K-means clustering	-	Prec = 0.78, Correlation Coefficient (CC)=0.95	Watershed transform requires speckle removal to improve SNR ratio.
43	Vidal et al. (2020) [149]	-Private (356 images)	-	HOG, GLCM, SVM, Intensity and texture descriptors	375	ACC-DRT-78.4% CME-84%, SRD-95.4%	-The visualization of the fluid regions is provided.
44	Mou et al. (2022) [153]	Mendeley(108312 images), Private (1021)	-	Bayesian NN, Epistemic uncertainty, Bayesian U-Net	-	ACC values, Mendeley = 94.9%, Private = 92.1%	-Existing anomaly detection approaches were outperformed by the suggested method.

Legend: Th = Thresholding, MF = Median Filtering, MIP = Morphological Image Processing, MSSP = Multi Scale Spatial Pyramid, LBP = Local Binary Patterns, SVM = Support Vector Machine, PCA = Principal Component Analysis, SNR = Signal to Noise Ratio, BM3D = Block Matching and 3D-filtering, RCF = Retinal Curvature Flattening, IC = Image Cropping, HOG = Histogram of Oriented Gradients, DiL = Dictionary Learning, BoW = Bag of Words, RF = Random Forest, NLM = Non-Local Means, GF = Gaussian Filter, NR = Noise Removal, LR = Logistic Regression, KR = Kernel Regression, GT = Graph Theory, DP = Dynamic Programming, SMO = Sequential Minimal Optimization, Re = Resize, ACC = Accuracy, SE = Sensitivity, SP = Specificity, Prec = Precision, AUC = Area Under the Curve, GC = Grayscale Conversion, ROI = Region of Interest, LCP = Linear Configuration Patterns, LR = Logistic Regression, NB = Naive Bayes, GB = Gradient Boosting, KNN = K-Nearest neighbour, GMM = Gaussian Mixture Model, ELM = Extreme Learning Machine, SURF = Speeded-Up Robust Features, NN = Neural Network, Jl = Jaccard Index, MT = Multilevel Thresholding, LS = Learning strategy, DC = Dice Coefficient, GLCM = Gray Level Co-occurrence Matrix.

Table 6 – Use of OCT images with DL models.

SN	Author & Year	Datasets & OCT Volumes	Techniques Used	Results	Observations
1	Lee et al. (2017) [154]	Private (1,289 images)	CNN to detect IRF/U-net autoencoder	Cross-validated DC = 0.91	-Training images had multiple lesions.
2	Awais et al. (2017) [189]	Private(32)	VGG16 KNN, RF classifiers	ACC = 87.5%, SE = 93.5%	-Multiple DL architectures could have been used.
3	Chan et al. (2017) [190]	Private(32)	AlexNet, BM3D filter	SE = 94%, SP = 97%	-Preprocessing enhances result.
4	Karri et al. (2018) [191]	Dukes(45)	GoogleNet, Classification of AMD, DME, and NORMAL	Mean-decision pooling DME-0.86	-Gradient explode/vanishing affects the result.
5	Rasti et al. (2018) [155]	Private(60)	Adaptive feature learning, Wavelet based CNN	Prec, Private-99.33%, Dukes-98.67%	-Eliminated the need for segmentation, de-noising.
6	Perdomo et al. (2018) [157]	Private(32)	End-to-end CNN, Cropping and resizing, Learned filters	ACC, SE, SP is 93.75%	-CNNs help improve results by learning data across layers.
7	Rasti et al. (2018) [69]	NEH(148) Dukes(45)	MCMC ensemble	Prec rate = 98.86% AUC = 0.9985	-Image denoising, retinal layer segmentations are eliminated.
8	Fauw et al. (2018) [222]	Private (14,884)	An ensemble of 5 segmentation, 5 classification models, 3D U-Net	For both the devices, AUC = 0.99	-Device-independent OCT segmentations.
9	Vahadane et al. (2018) [223]	Private(OCT scans from 328 patients)	Detection of fluid regions, Label prediction using CNN, Rule based method	Prec = 96.43%, Recall = 89.4% F1-score = 0.92	-Patches are beneficial over weak labeling, help reduce complexity.
10	Kaymak et al. (2018) [196]	Mendeley (84,484 images)	AlexNet Multi-class Classification	DME, SP = 99.6%, ACC = 99.8%, SE = 100%	-Demonstrates how training the system from scratch beats a TL-based model.
11	Chan et al. (2018) [203]	Private(32) Private(43)	AlexNet, VGGNet, GoogleNet PCA	DME, ACC = 93.75%	-No parameter optimization.
12	Schlegl et al. (2018) [156]	Private (1200)	IRF & SRD segmentation, Quantification, Encoder-decoder CNN	IRF, AUC-0.94 SRD, AUC-0.92	-Results can be improved with a larger SRF dataset.
13	Lu et al. (2018) [195]	Private (60407 images)	101-layer CNN-ResNet, TL, Four binary classifiers	For multi-class, AUC-0.984	-Model generalizability can be tested with large data sets.
14	Ji et al. (2018) [193]	Dukes(45) Private(1680 images)	Inception V3 CNN to extract the semantic level features	SP = 99.1% ACC = 98.8%	-On smaller medical data sets, the TL can perform better.
15	Kermany et al. (2018) [70]	Mendeley (207130 images)	Inception V3, TL, Occlusion Testing, Expert Comparisons	ACC = 96.6%, SE = 97.8% SP = 97.4%	-Method be used for high-impact therapeutic imaging applications.
16	Kamble et al. (2018) [194]	Private(32) Private(83)	Inception-Resnet-v2, Comparison with Resnet-50, Inception-V3	ACC, SP and SE is 100%	-Model obtained very good results.
17	Alsath et al. (2018) [221]	Optima(15)	Detection of retinal cysts, Autoencoder model	SE = 93.0% SP = 82.0%	-Inadequate results for patch labeling, 150 < threshold > 200.
18	Tennakoon et al. (2018) [164]	RETOUCH (70)	CNN, Skip-connections, Cross-entropy, dice and adversarial loss	Overall DC-0.75, Absolute Volume Difference-0.11	-Effective model to segment out fluids at voxel level.
19	Meng et al. (2018) [210]	Dukes(45)	Residual CNN, Less computation, 18-layer Residual CNN	Highest overall ACC obtained is 90%	-Recombined CNN with kernel size of 3x3 performs better.
20	Rashno et al. (2018) [160]	RETOUCH(112)	CNN model, Graph shortest path, Layer Segmentation, IRF and SRF	Topcon-0.821, Cirrus-0.764, Spectralis-0.922	-Good results(DC) for multi-vendor images.
21	Li et al. (2019) [170]	Private(75)	2D-FCNN, Data augmentation, 3D U-Net	ACC = 99%, F-score = 95%	-Eliminates impact of class imbalance.
22	Gao et al. (2019) [169]	Private(52)	Double-branched and Area-constraint FCNN, joint supervision learning	True Positive Volume	-Double-branched structure aids in learning multi-level OCT features.
23	Gopinath et al. (2019) [177]	Optima(80) Private(10) Dukes2(10)	Localizing cysts in 3-D OCT, Selective enhancement, CNN	Fraction = 99.57%, DC-0.933	-Model could not detect small cysts, confined to a particular slice.
24	Perdomo et al. (2019) [158]	Private(32) Private(43)	Extended version of OCT-NET [157], Volume pre-processing	Private-0.79	-Model evaluation on larger datasets is lacking.
25	Alqudah et al. (2019) [179]	Mendeley (108,312 images)	CNN, Multiclass classification, Softmax classifier	Multi-class, ACC = 98.1%	-Classifies 5 eye diseases, with largest set of images.
26	Bhowmik et al. (2019) [200]	Mendeley (84,495)	TL, VGG16, Inception, V3 SoftMax, activation	ACC, Training-99.9%, Testing-94%	-Semantic segmentation aids subtle changes within retina.
27	Moura et al. (2019) [204]	Private (400 images)	Deep feature extraction, AlexNet, VGG-19, Inception-V3	ACC = 99.94%	-TL reduces computation, performs well with limited data.
28	Li et al. (2019) [192]	Mendeley(207130 images)	TL, VGG-16, Statistical analysis	SE = 98%, SP = 99%	-TL improves ACC(98.6%).
29	Das et al. (2019) [178]	Mendeley (84,484 images)	Multi-scale Deep Feature Fusion, Cost sensitive loss function	ACC, SE-95.6% SP = 99.8%	-No manual fine-tuning of parameters, noise removal.
30	Fang et al. (2019) [159]	Mendeley (84,484 images)	Iterative Fusion CNN, Joint utilization of features of different convolutional layers	Overall ACC = 87.3%	-Increasing the feature maps beyond 64, will degrade the performance.
31	Li et al. (2019) [225]	Private (21,357 images)	Image labeling and augmentation, Multi-ResNet50 ensembling	SE = 96.3% SP = 98.5%	-Effective method for smaller datasets(ACC = 97.3%).
32	Fang et al. (2019) [235]	Mendeley(84,484 images) NEH(148)	Lesion-Aware CNN, LDN, Soft attention map	Model outperformed other methods on 2 datasets	-Lesion-related data combined with CNN can provide good results.
33	Mishra et al. (2019) [236]	Dukes(45) NEH(148)	Multi-Level Dual Attention Mechanism, Multi-Level Features Based Attention	AUC, Dukes-1, NEH-0.99 ACC, Dukes-99.9%	-No retinal flattening, time-consuming RoI selection, denoising.
34	Kamran et al. (2019) [161]	Mendeley(84,484 images) Dukes(45)	Residual unit, Atrous Separable Convolution, Gradient degradation prevention	ACC, Dukes-100% Mendeley-99.80%	-Addressed issues of gradient explode and gradient vanishing.
35	Vidal et al. (2019) [162]	Private (3247 images)	DenseNet, Pathological identifications, Intuitive color map	Mean AUC 0.9961 ± 0.0029	-Additional stability to over fitting with minimal datasets.

Table 6 – Continued.

SN	Author & Year	Datasets & OCT Volumes	Techniques Used	Results	Observations
36	Wang et al. (2019) [205]	Dukes(45) NEH(148)	Image Pretreatment ResNe50, DenseNet, Statistical Analysis	Both models obtained an ACC more than 98%, AUC=0.99	-TL can be employed with minimal datasets.
37	Grish et al. (2019) [166]	Optima(30)	CNN to segment CME, Depthwise separable convolutional filters	Model achieved a mean Dice score=0.74	-Vendor-independent IRC cyst segmentation.
38	Grish et al. (2019) [167]	Optima(30)	FCNN to segment CME, Device-independent method to counteract noise	Dice correlation coefficient=0.71	Hyperparameterization and optimal data augmentations prevent over fitting.
39	Sun et al. (2019) [176]	Dukes(45) Private(3000 images)	FCNN, Spatial pyramid matching based on sparse coding, Inception V3	Inception, ACC = 99.69%, SP = 99.77%	-No denoising, no layer segmentation, no manual parameter selection.
40	Hassan et al. (2019) [224]	Mendeley NEH(148)	Deep ensemble learning, CNN, Hybrid classifier, ANN, SVM, Naive Bayes	DME, ACC = 94.3% Mean DC=0.9019% ± 0.04	-Tested on multiple public datasets of different imaging modalities.
41	Lu et al. (2019) [174]	Mendeley(530 images) RETOUCH(70)	Multiclass fluid segmentation and detection, Graph-cut method, FCNN RF	Fluid, AUC=1.0, Segmentation Mean DC=0.766	-In 2017, the model took the lead in MICCAI RETOUCH challenge.
42	Hu et al. (2019) [172]	Private(100) [72]	Deep NN, ASPP, Stochastic ASPP	For SRF detection DSC = 0.8447	-Stochastic operation avoids validation errors, over fitting.
43	Liu et al. (2019) [237]	Optima (500 scans)	U-net based encoder-decoder, Attention Gate, Cross entropy, dice and regression loss	Mean DC = 0.829 Standard deviation = 0.016	-Joint-loss functions perform well than models trained with single loss.
44	Liu et al. (2019) [233]	Duke2(10), Private(10)	Semi-supervised U-net, Adversarial learning	DC, Duke2=0.90, Private=0.92	-Adversarial learning for segmentation.
45	Rasti et al. (2019) [226]	Dukes(45), NEH, Duke -Biopitigen(384) [238]	MCME, WCME	MCME Prec = 98.14% WCME, Prec = 96.06%	-WCME provides reliable method for classifying non-aligned OCT images.
46	Rong et al. (2019) [168]	Dukes(45) Private(7)	Surrogate-assisted model, Thresholding, Morphological dilation	Private, AUC=0.978 Dukes, AUC=0.985	-AUC of the ensemble technique is significantly higher.
47	Huang et al. (2019) [165]	Mendeley(84,484) Private(8904)	Layer Guided CNN, Segmentation network, Lesion-related layers	Public, OA = 88.4% Private, OA = 89.9%	-Association across lesions, certain retina layers, is used.
48	Rastogi et al. (2019) [198]	Mendeley (108,309 images)	DenseNet, Extracting Deep-features Determining ROI	Prec = 96.4%, AUC=0.99 SP = 99.15%	-Activation maps built to see if the model is focusing on specific region.
49	Chetoui et al. (2020) [199]	Mendeley(84484 images)	CNN EfficientNet, Global Average Pooling	ACC = 98.46% AUC=0.998	-Visualization is provided to understand the prediction.
50	Paul et al. (2020) [228]	Mendeley (84,484 images)	Deep Ensemble Network, Pre-trained architectures	Prec = 97.02% Recall = 97.50%	-Ensemble models yield better results(ACC = 98.53%).
51	Sun et al. (2020) [239]	Duke-Biopitigen(384) Private(910)	B-scan feature extractor, 2D feature map, ResNet-50, Volume-level classifier	ACC = 98%, SP = 95.65%, SE = 99.26%	-Image relabeling within volume must be incorporated to confirm results.
52	Azimi et al. (2020) [175]	Private (600 images)	FCN Model, Graph shortest path layer segmentation	Average DC = 0.86 SE = 87.38%	-Layer segmentation results affect fluid detection.
53	Hassan et al. (2020) [180]	Dukes(45)	25-layered CNN, SGD	ACC = 93.42% Mean DC up to 0.64	-Larger datasets may enhance result, increase training time.
54	Hwang et al. (2020) [209]	Private (4932 images)	Data set augmentation, VGG16, Inception V3	VGG, SE = 96.48% Inception, ACC = 93.09%	-Useful only for patients who are currently taking DME treatment.
55	Berrini et al. (2020) [208]	Mendeley (84,495 images)	Inception V3, VGG-16 Modified version of VGG-16	Inception V3 ACC = 99.27%	-Pre-trained models can improve ACC of findings.
56	Wang et al. (2020) [240]	Private1(1396) Private2(3248)	Weakly supervised learning, Uncertainty estimation, CNN, Bag-level prediction	Private1, AUC=0.99, Private2, AUC=0.98, ACC = 95.1%	-Model uncertainty estimation across different periods may affect results.
57	Ibrahim et al. (2020) [163]	Mendeley (108,312 images)	Hybrid CAD(HyCAD), DL, Traditional ML, Normal-VGG16, L2 kernel regularization	ACC = 98.8%, SE = 99.4% SP = 98.2%	Fusion of handcrafted, DL features improves the result.
58	Tsuiji et al. (2020) [182]	Mendeley (84,484 images)	Positional information, Capsule network, Visualizing feature maps	OA = 99.6%	-Capsule networks perform better in OCT classification.
59	Adel et al. (2020) [211]	Mendeley(84495 images)	TL, Inception V3, Xception	Xception, ACC = 98%	-TL, hinge loss as SVM performs well.
60	Seeböck et al. (2020) [171]	Private (148)	Bayesian U-Net, Epistemic uncertainty estimates, Monte Carlo dropout	Segmentation of binary masks of retinal anomalies, ACC = 95.7%, SE = 98%, SP = 100%	-Normal anatomical data can be used for anomaly detection.
61	Rajagopal et al. (2020) [183]	Mendeley (108,312 images)	Data augmentation, Kuan filtering, CNN, K-fold cross validation	ACC = 95.7%, SE = 98%, SP = 100%	-Model delivered superior ACC for a huge data set.
62	Wu et al. (2020) [241]	Mendeley(108309) Private(2564 images)	Attention-based method, Customized pre-processing, Multi-class classification	Multi-class, ACC = 97.4% Binary, SP = 100%	-Standalone and decoupled module with limited demand for training data.
63	Kim et al. (2020) [227]	Mendeley (108,309 images)	Ensemble of several CNNs, FCN to remove noise, Projection method	ResNet, ACC = 98.10% Ensemble, ACC = 98.9%	-They evaluated the performance of non-TL-based, TL methods.
64	Amaladevi et al. (2020) [206]	Dukes(45)	GoogleNet, VGGNet, AMD, DME, NORMAL classification	VGGNet, ACC = 96% SP = 99%	-In comparison to GoogleNet, VGGNet has performed better.
65	Vellakani et al. (2020) [242]	Mendeley (8,032 images)	Clinical decision support, LSTM, DenseNet201	OA = 96.9% Positive predictive value = 97.2%	-With DenseNet201, Xception image caption generators performed well.
66	Guo et al. (2020) [197]	Private(45)	CNN with U-Net, Multi-scale features	F1-Score = 0.864% ± 0.084	-Volumetric fluid segmentation.

Table 6 – Continued.

SN	Author & Year	Datasets & OCT Volumes	Techniques Used	Results	Observations
67	Rasti et al. (2020) [243]	Private (61 B Scans)	Treatment response prediction, Attention-to-DME, VGG	AUC-0.86, Prec = 85.5% SP = 85.0%	-Model predicts the response to anti-VEGF treatment in patients with DME.
68	Zhang et al. (2020) [201]	-Mendeley (38057 images)	Multiscale TL, Edge detection, InceptionV3	Prec = 97.2% SP = 97%	-Model can identify disorders even in poor quality images.
69	Suzuki et al. (2020) [202]	Mendeley(108309) NEH(148)	DL classifier, Data augmentation, Local Outlier Factor	Area under ROC = 1, SE, SP = 100%	-Only disease detection, no severity quantification.
70	Sanchez et al. (2020) [230]	Dukes2(384)	CNN ensemble, Modified Deep Retinal Understanding model	Fluid, DC-0.6245 HRF, DC-0.4437	-Model is able to quantify and segment the fluid present.
71	Das et al. (2020) [244]	NEH(148)	Data-efficient semi-supervised GAN	ACC = 97.43%, AUC-0.98	-No cost of image annotation.
72	He et al. (2020)	Mendeley(84484) Private(8904 images)	Label smoothing GAN Generator, Discriminator, Artificial pseudo labels	Mendeley, SP = 95% Prec = 87.25	-Model could augment the dataset and avoid overfitting with few images.
73	Padmasini et al. (2020) [207]	Private (376 images)	TL, AlexNet, ResNet 50, VGGNet	ResNet50, ACC = 99.3% F1-score = 97.5%	-7 primary lesions including CME, SRD, DRT were detected by the model.
74	Wei et al. (2020) [181]	Dukes2(110 images) Public(35) [245]	Mutex relationship among layers MDL, FCNN	On Dukes dataset Fluid segmentation, DC-0.81	-MDL enhances segmentation ACC substantially, for fluid class.
75	Alsaih et al. (2020) [229]	Dukes2(10), Optima(30) RETOUCH(112)	Semantic segmentation, Slice extraction, denoising, Deep CNN, BM3D filtering	RETOUCH, DSC-0.867 IRF, DSC-0.924	-Integration of data from many dimensions culminated in Accurate results.
76	Singh et al. (2021) [212]	Mendeley (85000 images)	Inception-v3 Comparison of 13 DL methods	Median rating-3.85/5, Deep Taylor-Rank 1.	-Model is able to highlight the structures for each disorder.
77	Wu et al. (2021) [213]	Mendeley(11599 images), Private(1547)	Classification of DRT, SRD, CME BM3D, Image multi-labeling	AUC, DRT-0.971 CME-0.974, SRD-0.994	-Demonstrated the efficiency of DL in the detection of SRD, CME, DRT.
78	Sunija et al. (2021) [186]	Mendeley(84484 images)	Deep CNN-6 convolutional blocks	ACC, SP = 99.69%	-Can be used for real-time detection.
79	Lin et al. (2021) [231]	Private(63000 images)	VGG-BN-16 CNN, RF ensembles	DME, AUROC-0.9735	-OCT Macula coverage affects results.
80	Mezni et al. (2021) [220]	Private(257 images)	Advanced filtering, BM3D, Gaussian noise removal, DBN	ACC = 92.54% SP = 93.25%	-Compared to existing methods, combined filtering, provides better results.
81	Wang et al. (2021) [185]	Private (8676 images)	ASPP, Fully connected conditional random field	AUC-0.96, Prec = 95.7% SP = 97.31%	-Significant improvement in results over hand-crafted, end-to-end methods.
82	Hassan et al. (2021) [215]	Mendeley(683), RETOU CH(112), OCTID(572)	RFS-Net, ASPP, Contracting module(CM), Residual, Inception module(IM)	F1-Scores, IRF-76.2% SRF-79.6%	-Elimination of segmentation offers large gain in results.
83	Liu et al. (2021) [232]	Mendeley(84484) RETOUCH(70)	Self-ensembling semi-supervised model, Uncertainty guidance, Distance map	RETOUCH, Mean DC-0.75 Aggregated JI-0.659	-Accurate detection of cystic boundaries similar to ground truth.
84	Liu et al. (2021) [246]	Mendeley(84484) Private(1650 images)	VGG16, Attention-based method, Bounding boxes, Attention block	Mendeley, OA = 90.4% Private, OA = 90.9%	-Data gained during classification is fed to segmentation task to improve results.
85	Liu et al. (2021) [247]	Optima(500 Scans)	Improved U-Net, Skip connections, Attention block, Multiple loss functions	DC, Cirrus-0.83, Nidek-0.80 Topcon-0.83, Spectralis-0.84	-Model can be used for OCT images obtained by different modalities.
86	Ahmed et al. (2021) [173]	Mendeley (84484 images)	BM3D filtering, Richardson-Lucy deconvolution	ACC = 99.81% DC-0.83, AUROC-0.99	-Model performs well even with images having noise, blurred edges.
87	Rajagopalan et al. (2021) [184]	Mendeley (84,000 images)	Novel CNN, Batch Normalization, Hyperparameters optimization, Feature visualization	ACC = 97.01% SP = 98.07%	-The model can be used to screen other eye disorders on a widespread scale.
88	Tuncer et al. (2021) [216]	Mendeley (968 images)	TL, CNN-SVM model, Alexnet, Resnet18, Googlenet	Highest ACC-GoogNet = 97.4%	-They incorporated SVM instead of softmax layer to obtain good results.
89	Sharma et al. (2021) [187]	Mendeley (84895 images)	Deep CNN, Random under-sampling, Dropout layer for regularization	ACC = 99.38% Prec = 99.38% AUC-1.00	-Dataset was oversampled, with various filters to address class imbalance.
90	Wang et al. (2021) [214]	Dukes(45)	TL, Pretrained VGGNet-16 Evaluation indicators	ACC = 97.9% Sensitivity = 98.0%	-Severity and type of DME detection could be incorporated.
91	Pranoto et al. (2021) [218]	Mendeley (37663 images)	TL, VGGNet-16, MobileNet Custom CNN	Best results for custom CNN ACC = 96%	-All models suffer from over fitting due to smaller training dataset.
92	Hassan et al. (2021) [188]	Dukes, Dukes2, BIOMISA Mendeley(43613 scans)	Hybrid CNN, Lesion-influenced grading, Classification of multi-vendor scans	Lesion extraction mean IoU score-0.805, F1-score = 99.5%	-Model can extract multiple lesions in OCT scans from multiple vendors.
93	Upadhyay et al. (2021) [217]	Mendeley (84895 images)	Coherent CNN, Concept of binding, VGG-16	ACC = 97.19%	-Network layers were cohesively linked to allow for a smooth flow of image features.
94	Smitha et al. (2022) [248]	Mendeley(83605 images), NEH(148)	GAN-based approach, Gabor features	ACC = 92.42% F1-score = 79%	-Only RPE and ILM layers were clearly apparent in the fake images.
95	Togacar et al. (2022) [219]	Mendeley(84484) Dukes(45), NEH(148)	Class-based activations, Slime mold optimization, Nine TL models	Accuracies, Duke = 99.89% Mendeley = 99.6%, NEH = 97.5%	-Among the activation sets acquired by CNN models, the model could pick the dominant activations.

Legend: CNN = Convolutional Neural Network, IRF = Intra Retinal Fluid, DC = Dice Coefficient, RF = Random Forest, BM3D = Block Matching and 3D-filtering, AMD = Age related Macular Degeneration, NEH = Noor Eye Hospital, SVM = Support Vector Machine, PCA = Principal Component Analysis, SRD = Serous Retinal Detachment, CME = Cystoid Macular Edema, DRT = Diffuse Retinal Detachment, SRF = Sub Retinal Fluid, FCNN = Fully Convolutional Neural Network, DBN = Deep Belief Network, ANN = Artificial Neural Network, ASPP = Atrous Spatial Pyramid Pooling, MDL = Mutex Dice Loss, MCME = Multiscale Convolutional Mixture of Experts, LSTM = Long Short-Term Memory networks, WCME = Wavelet Convolutional Mixture of Experts, TL = Transfer Learning, LDN = Lesion Detection Network, SGD = Stochastic Gradient Descent, DL = Deep Learning, VGG = Visual Geometry Group, KNN = K-Nearest neighbour, NB = Naive Bayes, CLAHE = Contrast Limited Adaptive Histogram Equalization, FRCNN = Fast Region-based Convolutional Neural Network, GAN = Generative Adversarial Network, RPE = Retinal Pigment Epithelium, ILM = Inner Limiting Membrane, ACC = Accuracy, SE = Sensitivity, SP = Specificity, Prec = Precision, AUC = Area Under the Curve, DME = Diabetic Macular Edema, DR = Diabetic Retinopathy, ROC- Receiver Operating Characteristics, OCT = Optical Coherence Tomography, CST = Central Subfield Thickness, CFT = Central Foveal Thickness, HE = Hard Exudate, DED = Diabetic Eye Disease, OD = Optic Disc, DLS = Deep Learning System.

2.2.1. Traditional ML-based MTIF approaches

Traditional ML-based techniques for extracting features from retinal OCT images consisted of Multi-Scale Spatial Pyramids (MSSP), Local Binary Pattern (LBP) histograms, bilateral filters, Histogram of Oriented Gradient (HOG), Linear Configuration Patterns (LCP), Speeded-Up Robust Features (SURF) descriptors, dictionary learning, etc. A vast number of articles listed in this section have used these methods for extracting features for the classification of multiple eye diseases such as DME, MH, AMD and CNV. Liu et al. [115] employed these methods to study the effect of the size of the training dataset on results obtained. A relatively fast bilateral filter enabling speckle denoising while maintaining CME margins is included in a CME segmentation and quantification model [116]. Srinivasan et al. [66] developed a method for distinguishing DME, AMD, and NORMAL by utilizing several B scans from an OCT volume. The proposed strategy avoids the overhead of inner retinal layer segmentation. In [117], volume profiles produced using LBP texture provide a significant discriminative ability for identifying DME and NORMAL OCT volumes. The impact of various pre-processing techniques, the influence of feature extraction and mapping algorithms, and the effect of the codebook size in BoW along with an assessment of several strategies for classification have all been investigated in depth by Lemaitre et al. [118]. The multi-scaling pyramid setting, LCP features along with the significantly reduced number of features yielded good performance for DME and AMD detection by Wang et al. [119]. Through the assistance of a pipeline of conventional ML-based techniques, Alsaih et al. [120] offered a mechanism for automatically classifying DME and NORMAL volumes. In order to extract the pathology of cysts, authors Arif et al. [121] created a retinal mask and a retinal thickness profile, which is used to investigate retinal thickness variations in DME patients. Sun et al. [122] described a method for distinguishing DME, NORMAL and AMD using cropping and aligning OCT scans to build global representations.

Athira et al. [123] derived the feature vector using segmentation-based fractal texture analysis to classify ME images. Dynamically quantified clinical-driven hand-crafted features of DME, AMD and NORMAL subjects were used with several classifiers with random forest yielding the best results [124]. Dash et al. [125] pre-processed the OCT scans using image processing methods followed by segmentation of six retinal layers and estimation of the area between every two layers to classify OCT scans. To speed up the extraction of features, they used Gaussian square-shaped filters and blob detectors relying on the determinant of the Hessian matrix [126]. Mousavi et al. [127] used three types of dictionary learning to identify common and distinct patterns in OCT datasets for eye disease classification. Iterative high-pass filtering approach is employed to segment out layers and detect cysts within outer nuclear regions, inner nuclear, and inner plexiform regions by Roychowdhury et al. [128]. Segmentation of various retinal layers was carried out to detect various edema types and multiple eye disorders. Sugruk et al. [129] employed segmentation of Retinal Pigment Epithelium (RPE) and retinal nerve fibre layer complex for the classification of AMD and DME. Syed et al.

[130] extracted retinal layers from OCT images using structure tensors.

The automated method to classify ME, CSR and NORMAL by Hassan et al. [131] was based on the segmentation of choroid and Inner Limiting Membrane (ILM) obtained from the highly coherent 2D tensor corresponding to the candidate image. The authors further demonstrate a completely automated model based on SVM to identify maculopathy using coherent tensors and features extracted from the retinal fluids and thickness profiles [132]. Few authors have demonstrated cyst detection and quantification methods based on regression models along with graph theory and the dynamic programming [71]. Hernandez-Martinez et al. [133] assessed presence of IRF in conjunction with CFT to detect DME using a binary logistic regression model. Coherent tensors were employed in the identification of DME by authors Naz et al. [134]. To identify the retina layers, Maurya et al. [135] exploited gradient information from OCT pictures. CME and SRD were detected using hyporeflexive regions within the layers along with other clinical features.

The method based on marker-controlled watershed transform proposed by Girish et al. [136] outperformed the state-of-art cyst segmentation methods. Sidibe et al. [137] modeled the appearance of OCT normal B scans using a GMM to identify abnormal images as outliers. The method is unique in the sense that it does not require a manually annotated training set. Fang et al. [138] presented a completely automated technique to reap the benefits of the high correlation within features in 3-D image data. Girish et al. [139] steered a benchmarking study focused on assessing various methods for the segmentation of intra-retinal cysts. Chen et al. [140] developed a method to estimate CME, using gamma transformation, which eliminates pixel intensity variability and hence minimizes false detection. As DRT edema originates in the outer retina, a variety of 307 features from that region have been used to recognize and segment DRT by Samagaio et al. [141].

The fluid detection method employing neutrosophic sets does not require human seed points and was robust against fluctuation of the intensity values for fluid regions in OCT scans [142]. The proactive approach for characterizing the presence of SRD edema by Moura et al. [20] includes several intuitive visualization modules, comprising pertinent indicative biomarkers, to enable practitioners in their diagnostic and treatment procedures. They also presented an approach for extracting retinal vessels that allow for precise segmentation of DRT edemas [143] along with detection of all three edema types [144]. In [145], authors defined two regions of the retina: one corresponding to the ILM/Outer Plexiform Layer (OPL) region and another to the OPL/RPE region. The method then identifies where exactly inside these retinal regions each form of DME is prevalent. In [146], authors extracted DME biomarkers, utilizing numerous image processing and ML approaches to detect the existence of SRD, CME, and DRT along with Epiretinal Membrane (ERM) in DME patients. Wu et al. [147] developed a framework for detecting IRF and DR, based on level-set and Gaussian mixture model (GMM) clustering. Various fully automatic approaches for simultaneous identification of DRT, CME, and SRD were adopted in the literature some of which

attempt to quantify them [148]. In [149], an effective identification and visualization methodology for DME types based on the examination of distinct image regions is proposed by the authors. In a dataset of 356 labelled images, they investigate a thorough and diverse library of 375 texture and intensity properties. In [150], authors first define four major retinal areas that signify the outer retina, which is where DRT edema often manifests. Then from the regions, a vast and diverse set of features were derived and fed to various ML classifiers. In another approach, to identify each subtype, the authors used distinct adaptive thresholding techniques and learning strategies along with clinical attributes [19]. Otero et al. [151] presented an automatic method by using regional analysis and also an intuitive colour map is built to detect the simultaneous identification of DME types. Girish et al. [152] combined conventional techniques with cutting-edge unsupervised algorithms for cyst segmentation. The method used by Mou et al. [153] was based on anomaly detection and uses uncertainty estimation to help resolve the challenges of data scarcity.

2.2.2. DL-based approaches

DL approaches for DME detection using OCT images used in the literature include supervised approaches such as customized CNN and TL models, Ensemble of DL classifiers, GAN-based methods, Deep Belief Networks, Recurrent Neural Networks (RNN), and autoencoder based unsupervised methods.

2.2.2.1. Custom CNN models. Lee et al. [154] developed a CNN to detect IRF in OCT images by incorporating pixel-wise probability generation. Macular images which were manually segmented were used for training. To examine the intraretinal layers, anomalies or lesion patterns, the framework by Rasti et al. [155] extracted cumulative features of 3D OCT volumes with the help of a wavelet-based CNN hence, eliminating segmentation and retinal alignment. A semantic segmentation technique to segment out the fluid regions was proposed with pixel-wise labelling of the OCT image [156]. Perdomo et al. [157] introduced a new end-to-end classifier based on CNNs that can accurately detect important DME patterns from OCT scans. They also offered a feedback stage to highlight the areas in the scans which may help clinicians in assessing the model prediction [158]. Fang et al. [159] adopted an iterative fusion CNN to successfully collect and use the information from all the convolutional layers for the automatic classification of OCT images.

Rashno et al. [160] demonstrated a fully automated approach for segmenting three types of fluid in an OCT volume, including IRF, SRF, and PED. For layer segmentation and flattening, the method utilizes graph shortest path techniques, and CNN for classification. Kamran et al. [161] proposed a CNN model that is trained from scratch to include a new residual unit and a building block along with a novel mechanism to eliminate the gradient vanishing problem. Many researchers have experimented with CNN models that were trained from scratch with sufficient training samples [162]. Ibrahim et al. [163] used a deep efficient feature fusion phase where features extracted from the ROI phase were combined with OCT image features extracted by Deep CNN.

The reported method by Tennakoon et al. [164] relies on a customized U-Net architecture that's been trained with a blended loss function that had an adversarial loss component. Huang et al. [165] employed a model where the layer segmentation maps and probability maps related to lesions were created and resulting information is utilized by layer guided CNN. With the use of depthwise separable convolutional filters, Girish et al. [166] designed a method to detect cysts in OCT images with substantially lowered computational parameters. They further demonstrated a manufacturer-independent CME area segmentation algorithm relying on the FCN model [167].

The input images, denoised along with their masks were then leveraged to create a large number of surrogate images for training the CNN by Rong et al. [168]. Gao et al. [169] developed a unique FCN based on area constraint and double-branched structure to learn low and high-level representative features. Li et al. [170] proposed a 3D FCN to segment out IRF and SRF and to eliminate the class imbalance issues in the dataset by modifying the loss function and network parameters. Seebock et al. [171] employed a Bayesian U-NET trained on normal images, and estimated uncertainty during testing of the images. Various reported methods made use of customized U-Net models for accurate segmentation of retina layers and fluid types. SRF and PED lesions in OCT images were segmented using an encoder-decoder model and atrous spatial pyramid pooling by Hu et al. [172]. Ahmed et al. [173] tuned the hyperparameters of a U-net model to detect and quantify CME. Lu et al. [174], performed multiclass segmentation of fluid pixels by grouping them into non-fluid areas (background), SRF, IRF and PED using an FCN. In [175], authors extracted the first and last layer of the retina to delimit the ROI. An FCN with two input channels: the OCT scan and its related ROI was presented to segment the fluid and the tissue areas. In [176], segmentation accuracy obtained using an FCN is validated and its performance in recognizing diseased macula in OCT images is evaluated and compared with conventional ML classifiers. Gopinath et al. [177] proposed an automatic technique for cyst segmentation that is generic and independent of modalities and therefore does not necessitate layer segmentation. Several studies used custom-built CNNs to classify a multitude of eye disorders like DME, AMD, drusen, CNV [178]. More accuracy and lower processing times were achieved by fine-tuning the CNN by employing the adam optimization approach for classifying multiple eye diseases [179]. Hassan et al. [180] used a 25-layer CNN to diagnose every B scan in the processed OCT volume from the pre-processing step for classifying ME, AMD, CSR, and NORMAL by extracting the fluid information. Wei et al. [181] presented an end-to-end FCNN based on depth max-pooling (DMP), with a distinct loss function that incorporated the mutex relationship between various retinal layers.

A CNN with a capsule network helps to learn and incorporate positional information from images [182]. Rajagopalan et al. proposed deep CNNs for multiple eye disease classification which employed a Kuan filter for despeckling of the OCT images along with a K-fold validation method to ensure proper dataset utilization [183]. They also designed a CNN by tuning the hyperparameters using a random search optimization method [184]. A new technique was presented to

segment ME in OCT, including pre-processing of image data through the use of speckle de-noising along with coarse segmentation of ME based on Atrous Spatial Pyramid Pooling (ASPP) by Wang et al. [185]. Sunija et al. [186] designed a CNN with reduced parameters so that it becomes easy to deploy the model in hospital scenarios. The model uses gradient-based class activation maps to gain an understanding of the predictions. Sharma et al. [187] presented a CNN with a dropout layer to prevent overfitting. Hassan et al. [188] presented a method for comprehensive retina assessment with a CNN that retrieves numerous retinal lesions.

2.2.2.2. TL-based models. TL has been used extensively by a vast number of researchers for the automated classification of OCT images. Features were extracted at multiple network layers including before the fully connected layer and after each fully connected layer using VGG16 by Awais et al. [189]. After extracting OCT scans from their volumes, they were pre-processed and concatenated thrice after which, feature extraction was carried out using AlexNet. The extracted features were fed to SVM for categorization in addition to 8-fold cross-validation [190]. To increase its prediction capability and to understand learned filter attributes by recognizing notable responses during prediction, Karri et al. [191] fine-tuned GoogleNet. Very good results have been obtained by training the VGG-16 network with 207,130 retinal OCT images and evaluating the results on a validation set [192]. TL was used by Kermany et al. [70] to train a neural network using a small fraction of the data required by conventional methods. They additionally offered a more clear and understandable diagnosis by emphasising the areas that the neural network had identified. The last few layers of the pre-trained Inception V3 model were discarded by Ji et al. [193], who interpret the remaining network as a fixed feature extractor during DME classification. In comparison to conventional learning, Kamble et al. [194] fine-tuned the Inception-Resnet-v2 model to effectively identify pathologies. In order to train and optimize the model for OCT classification, Lu et al. [195], employed OCT pictures labelled by 17 certified retinal specialists. The model was a 101-layer ResNet with 10-fold cross-validation technique. Several authors have employed the use of pre-trained AlexNet [196], ResNet [197], DenseNet [198] and EfficientNET-B7 [199] models for DME classification. By dropping out the contribution of a few randomly chosen neurons during model training, the authors of [200] demonstrated the effectiveness of dropout layers employed for regularisation. The research by Zhang et al. [201] aims to offer a high-speed and effective multi scale TL model for medical image processing to accomplish accurate and efficient disease classification. Suzuki et al. [202] demonstrated that on 108,308 retinal OCT pictures, the DL-based binary classifier with four fold cross-validation obtained very good classification results. Multiple TL models have been used by a few authors [203](AlexNet, VGG and GoogleNet), [204] (AlexNet, VGG and Inception-V3), [205](ResNet, DenseNet), [206] (VGG, GoogleNet), [207] (AlexNet, VGGNet, and ResNet-50). Berrimi et al. [208] proposed a CNN model to classify OCT images and they also compared the results with pre-trained models with modification by introducing regularization. Data augmentation methods such as random shear, zoom, rotation

and horizontal flip were used by the authors employing pre-trained models (InceptionV3 and VGG16) with cross-validation for DME classification [209]. To detect multiple classes, Meng et al. [210] offered a recombined residual CNN with less computation. Moura et al. [204] integrated deep features and features selected through several traditional strategies to obtain the most relative features.

Adel et al. [211] used a categorical hinge loss as SVM instances and two TL models for multi-disease classification. Singh et al. [212] carried out one of the pioneering investigations to compare the quality of several AI methods that used a broad panel of experts. Wu et al. [213] attempted the detection of three edema types along with their visualization through an occlusion test. Wang et al. [214] fine-tuned the VGG-16 network without any user intervention along with a softmax classifier to automatically identify and extract the features of DME. Hassan et al. [215] developed a model to obtain better features and save additional global data for accurate segmentation of retinal fluid. Tuncer et al. [216] proposed a hybrid CNN-SVM model for the classification of OCT which does not require additional feature extraction or noise filtering. In [217], authors profoundly detected irregular patterns in each class of retinal images by using a cohesively bound, 5 layered model with a batch normalization layer. Pramoto et al. [218] used several TL models along with a custom CNN model with 5 layers to classify DME. In [219], authors employed nine TL models, to retrieve class-based activations for each model using a slime mold algorithm to obtain the dominant activations.

2.2.2.3. Deep Belief Network (DBN) based methods. DBN is a type of Deep Neural Network comprised of layers of Restricted Boltzmann Machines stacked on top of each other. A DBN can learn to rebuild its inputs based on probabilities. Mezni et al. [220] designed an image classification method to classify OCT subjects into NORMAL, DME, and AMD using a DBN.

2.2.2.4. AE based methods. To label the cyst regions, the method by Alsaih et al. [221] utilizes a generalized classification pipeline that selects the stable areas and then correlates them to the ideal ground truth. The autoencoder then extracts features from the candidate regions in an unsupervised way.

2.2.2.5. Ensemble of DL classifiers. Rasti et al. [69] used a model for retinal disease diagnosis that does not rely on image denoising, complete retina layer delineation. The results obtained for an ensemble model [222] showed that the more the number of segmentation and classification instances, the better the performance. Vahadane et al. [223] used traditional image processing to detect candidate patches for fluid in OCT. Further, a CNN was used to predict the labels for candidate patches along with a rule-based method. Features from fundus and OCT images were extracted using a CNN and were fed to a supervised ML classifier ensemble [224]. To classify OCT pictures, the technique by Li et al. [225] used an ensemble of four classification models. Rasti et al. [226] developed an ensemble model that can work on OCT images without de-noising and alignment. Kim et al.

[227], reported good results for ResNet152, among the several TL models. They used an FCN to irradiate noise from the input OCT images. Paul et al. [228] used an ensemble of several TL models and a custom CNN to classify eye disorders. Alsaih et al. [229] have examined ten DL networks which have been ensembled in various dimensionalities. Sanchez et al. [230] presented a computational ensemble of two CNNs that automatically detected fluid and HRF. Lin et al. [231] developed an ensemble that empirically revealed better classification results as a consequence of increased macular coverage. A semi-supervised, uncertainty-guided self-ensembling model is proposed by Liu et al. [232], which consists of a teacher model and a learner model assisting in the diagnosis of DME.

2.2.2.6. GAN based methods. An ophthalmologist's annotation of OCT images is time-consuming and biased, and the number of labeled retinal OCT data is rather limited. In [233], authors modified U-net to have a large kernel size with a larger receptive field that can effectively use labelled and unlabelled data for OCT image segmentation. The segmentation model was trained in a semi-supervised, adversarial fashion. A novel automated GAN-based classifier was assessed for use in detecting progressive retinal disorders.

The model exhibited superior generalizability, even under highly restricted labelled data availability [244]. In [249], a classifier was trained with both synthetic unlabelled images and real labelled images to improve the OCT classification results with GANs. In [248], authors used GAN with the integration of manually extracted Gabor features to optimize the segmentation of retina layers in addition to non-local denoising to eliminate noise.

2.2.2.7. RNN based methods. RNNs are among the most promising algorithms in use as they are the only neural networks with internal memory. Wang et al. [240] combined the uncertainty estimation method with a multiple instance learning (MIL) strategy to build a strong instance classifier. Different deep CNN models and Long Short-Term Memory networks (LSTM) were utilized by Vellakani et al. [242] to build the multi-class classification system.

2.2.2.8. CNN models with AM. Due to the obvious bottleneck in information processing, human beings choose to focus on specific information while dismissing other apparent knowledge, as per cognitive science. And this is often known as an attention mechanism. In the past few years, attention methods have indeed been extensively exploited in image classification task [247]. A method by Liu et al. [237] makes use of U-net, as well as an attention structure to dynamically target the fluid regions. Fang et al. [235] employed a lesion-aware CNN to classify images by generating a soft attention map. Mishra et al. [236] employed a CNN where attention based on multi-level features and the self-attention mechanism of the finer features is considered. The focus of the research by Rasti et al. [243] was to modify the VGG network to incorporate attention using thickness-aware and self-attention mechanisms for DME detection. Based upon the 2-D feature map and CNN with/without attention mechanism, Sun et al. [239] presented a computer-

ized diagnosis of macular disorders in OCT volumes. Wu et al. [241] constructed an attention model that was integrated into a standard DL network and provides extra attention to sensitive regions having anomalies. In the case of conventional CNNs, the information learned during the classification process cannot be transferred back to the previous tasks such as segmentation. In order to make two tasks promote one another, Liu et al. [246] proposed a unique one-stage attention-based approach for weakly supervised lesion segmentation and multi-label image categorization. In [247], authors further developed a novel architecture with an attention block, for the categorization and segmentation of retinal OCT images to avoid extensive computation with multi-stage techniques.

3. General observations and discussion

This section provides a useful summary of the publications reviewed along with the general observations corresponding to two imaging modalities and two types of methodologies adopted by the different authors. The presented review is novel in that it explores several aspects of DME detection that have never been addressed in earlier reviews of the domain. A comparative analysis of the proposed review with other few relevant review publications are depicted in Table 7.

There have been similarities and differences between the DME detection techniques used for CFP and OCT image classification in the literature. The Fig. 6 shows similarities and differences between various state-of-the-art DL and ML DME detection techniques used for CFP and OCT image classification. The DME detection techniques used for CFP and OCT picture classification are indicated by two ellipses in light green colour. The area of intersection of the ellipses in green, (indicated by ellipses in yellow) shows the similarity in the techniques used in the literature. The ellipses filled with pink colour illustrate the ML-based and DL-based methods. MTIF-based methods have been used for both CFP and OCT image classification. Fig. 7 shows the similarities and differences in the MTIF-based approaches shown in Fig. 6. Blue rectangles exhibit similar techniques, however, green rectangles illustrate differences. The advantages and limitations of CFP, OCT, ML-based methods, and DL-based methods are discussed along with a set of future research directions that can be investigated by the interested researchers. The dynamics of developments of the presented methods in the literature too are discussed and elucidated.

3.1. CFP for DME detection

Tables 3 and 4 elaborate on 75 articles in the reviewed literature elucidating the widespread use of fundus images for DME detection. The usage of CFP is restricted to the recognition of lesions that are located within the surface of the retina, for instance, haemorrhages, microaneurysms, and exudates (both hard and soft). As a consequence, CFP is a wiser alternative for DR diagnosis than DME. Because of poor depth resolution and lack of quantitative data, one cannot acquire information about the cross-sectional morphology of the retina with its thickness profile [137]. Fundus photography

Table 7 – Comparison of the proposed review to the relevant state of art reviews.

Reference	Fundus images Included	OCT Images Included	Public Fundus Datasets Listed	Public OCT Datasets Listed	ML methods with Fundus	DL Methods with Fundus	ML methods with OCT	DL methods with OCT	Number of papers	Time period considered	Disease Types
Mookiah et al. [46]	Yes	Yes	—	—	Yes	—	—	—	88	Not Reported	DME
Sarki et al. [103]	Yes	—	—	—	—	Yes	—	—	65	2014–2020	DR, DME, Cataract, Glaucoma
Rajesh et al. [256]	Yes	—	Yes	—	Yes	—	—	—	13	Not Reported	Diabetic Maculopathy
Sengupta et al. [257]	Yes	—	Yes	—	—	Yes	—	—	62	2014–2018	DR, DME, AMD Glaucoma
Li et al. [258]	Yes	—	Yes	—	—	Yes	—	—	143	January 2016– August 2020	Multiple Eye Diseases
Asiri et al. [259]	Yes	—	Yes	—	—	Yes	—	—	Nearly 87	Not Reported	DR
Badar et al. [260]	Yes	—	Yes	—	—	Yes	—	—	34	Articles till December 2018	Multiple Eye Diseases
Ajaz et al. [255]	Yes	Yes	—	—	Included DME detection models, No categorization of ML and DL based methods with subcategories	—	—	—	96	2000–May 2021	DME
Proposed Review	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	214	2010–October 2022	DME

Legend: DL = Deep Learning, ML = Machine Learning, OCT = Optical Coherence Tomography, DME = Diabetic Macular Edema, DR = Diabetic Retinopathy, AMD = Age related Macular Degeneration.

creates 2D photographs that assist in qualitative assessment rather than quantitative disease evaluation. Also, since images from seven different fields need to be collected, fundus imaging modality is time-consuming and tedious [250]. The mydriatic drops used for dilation of the pupil cause discomfort for patients [251]. Due to the obvious variability of the images in terms of retinal morphological characteristics, color, and indeed the presence of lesions, analyzing a retinal fundus image becomes a tough process. Addressing all types of lesions along with early clinical signs of DME present within the layers of the retina is not possible [54]. The researchers' analysis of the literature implies that this area is quite well established, and it is difficult to pinpoint future research opportunities in the detection of DME using fundus images with ML-based techniques.

3.2. OCT for DME detection

Tables 5 and 6 elaborate on 139 articles in the reviewed literature elucidating the widespread use of OCT images for DME detection. OCT offers easy-to-use, cost-effective, and patient-friendly imaging technology to diagnose various diseases with improvement in acquisition speed, sensitivity, and depth-resolved volumetric images. OCT is often used to examine the retina to make a diagnosis and assess a multitude of ocular abnormalities, especially DME, glaucoma, AMD, and CNV. OCT scans help in the evaluation of morphological features including the form and density of drusen, cysts, MH, and blood vessels. As a result, processing OCT pictures seem critical for large-scale investigations of abnormalities inside the retinal layers [17]. The number of publications based on disease/edema type classification is depicted in Fig. 8. Identification of CME, DRT and SRD seems critical in DME diagnosis as they differ substantially in terms of heterogeneity in their appearance [144,145,150]. 50 publications were aimed at the identification of CME by using multilevel thresholding algorithms, clinical information, and segmentation strategies whereas 36 papers aimed at the detection of SRD by using intensity and texture features, and segmentation techniques with very few using DL methods. Very few publications attempted the detection of DRT edemas using ML-based techniques, as they have complex and fuzzy appearance patterns. There is a need to develop sophisticated algorithms which can help in the accurate detection of DRT edema. Only 8 articles incorporated the detection of all the three morphological variants of DME using private datasets. As these edema types can co-exist with each other, there is a need to develop advanced algorithms to locate them with accuracy.

As a consequence of models leveraging proprietary datasets, there is no way to evaluate the efficacy of a newly built model for edema type detection. With regards to the comparative evaluation, researchers ought to remember that no other dataset has indeed been annotated with three kinds of ME occurring simultaneously [144]. OCT enables ophthalmologists to recognize specific biomarkers that aid in pathology identification sooner and forecast the outcome of treatment in DME. There are few, novel OCT imaging biomarkers outlined in the literature which helps prognosticate patients with DME. The prominent among these include

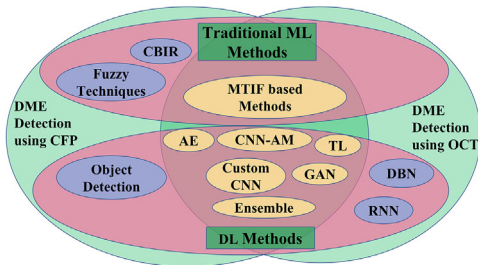


Fig. 6 – Similarities and differences in various techniques used in the literature.

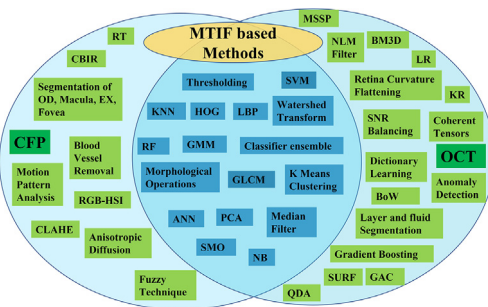


Fig. 7 – Similarities and differences in the MTIF-based techniques employed for CFP and OCT pictures.

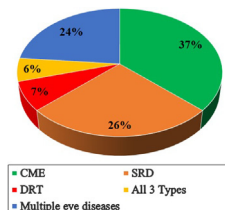


Fig. 8 – Illustration of methods based on edema type identification.

Disorganisation of Retinal Inner Layers (DRIL), Hyper-reflective Retinal Foci (HRF), HEs, the intactness of Ellipsoid Zone (EZ) in addition to External Limiting Membrane (ELM) [252]. These biological markers help identify patients at low risk by predicting the disease progression to the vision-threatening stage [253].

3.2.1. Layer segmentation approaches used in the literature

A very crucial step in identifying the various morphological subtypes of DME and the associated biomarkers is the segmentation of the retinal layers. They help us define the exact boundaries of the retinal regions where the different forms of ME and biomarkers typically manifest. The numerous retinal layers must be appropriately localised, which takes considerable experience and knowledge. Fluid segmentation into IRF and SRF is crucial for determining the type of edema. Therefore, it is imperative to have effective algorithms for differentiating the retinal layers. Annotating the retinal layer boundaries manually is time-consuming and subjective. This

has sparked a great deal of research into devising advanced algorithms for segmenting distinct retinal layers from OCT images, which can assist in an accurate diagnosis of eye diseases. Fig. 9 depicts the various retina layer and fluid segmentation methods used in the literature. Compared to other methods for layer segmentation, U-Net-based methods could obtain better results (Mean DC-0.900) for layer and fluid segmentation [233]. U-Net model with efficient fine-tuning of parameters can yield good model accuracy of 99.81% [173]. Attention-based U-Net model with tailored joint loss functions and skip-connections can provide good results for fluid segmentation [247].

3.3. Dynamics of development of the ML methods

Various ML-based methodologies used by different authors in the classification of retinal images is depicted in Tables 3 and 5. Initial research in the domain of DME detection started with traditional ML strategies for feature extraction including HOG, scale-invariant feature transform, dictionary learning using BoW, LBP, MSSP, GLCM, LCP, etc. The features extracted from CFP and OCT images using various methods were then fed to different ML classifiers, some of which include SVM, Naive Bayes, RF, and NN. As SVM provided the best classification results, it was used by a multitude of models in the literature with various feature extraction methods. The number of journal papers utilizing the various classifiers is illustrated in Fig. 10. CBIR systems used in the classification of CFPs performed satisfactorily as they depended on the extraction of low-level features. Further, when CBIR models exploited wavelet-based features, a decent improvement in the performance was observed. Subsequent CFP image classification models employed the fuzzy analysis that assisted in the design of more precise diagnostic systems [25,26]. The MTIF-Based Methods attempted the manual extraction of features and segmentation of lesions which seemed tedious and they provided satisfactory results. Later on, attempts were made to combine features from several feature extractors and use an ensemble of classifiers to enhance prediction. Later advancements led to the usage of ANNs, which appeared to be a promising solution to enhance outcomes.

Research on OCT images for DME detection started in the year 2011, with traditional ML methods such as LBP, MSSP for feature extraction, and SVM for classification to offer excellent outcomes (AUC-0.985) [115]. The discriminative hand-crafted features like HOG along with SVM offered good performance for DME detection (Accuracy = 100%) [66]. Instead of considering a single B scan for a subject, few models employed the examination of the entire OCT volume for improving the prediction.

Further developments in the field led to the segmentation of retinal layers using graph-based, gradient, and threshold-based methods. Identification of retinal layers helped in the detection of DME lesions located within these layers. Recent developments in the field have led to the segmentation of various edema types such as DRT, CME, and SRD. Numerous researchers used ML methods to identify the co-existence of all three forms of edema. Classical ML approaches are easy to understand and interpret because of the manual feature engineering involved. Moreover, because of a better grasp of

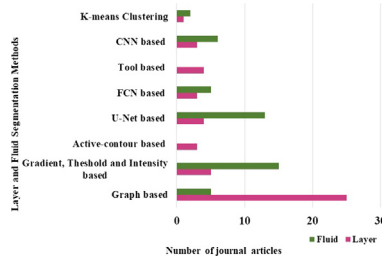


Fig. 9 – Techniques used for segmentation of retinal layers and fluid in OCT.

the data and underlying algorithms, adjusting hyperparameters and altering model designs are easier. A huge training data set is not necessary for handcrafted descriptions. Handcrafted features possess limitations in that the features obtained are often heavily dependent on the experience and knowledge of the designer and may not be effective. As a result, numerous researchers agree that DL methods are preferable for diagnosing DME and propose the use of DL for future research [144,149].

3.4. Dynamics of development of the DL methods

Various DL-based methodologies used by different authors in the classification of retinal images are depicted in Tables 4 and 6. In the realm of ophthalmology, in general, as well as for the diagnosis and management of ME, DL has made tremendous progress [254,255]. Fig. 11 depicts the number of journal articles that use various DL techniques.

Initial research in the automated detection of DME and DR using CFPs employed the use of Inception V3 which was trained on the ImageNet dataset [54]. Majority of TL models employed pre-initialization of networks using pre-trained weights along with batch normalization to avoid training the model from scratch. Later research in the field aimed at the profound use of other TL networks such as AlexNet [100,196], VGG16 [192,214], GoogleNet [191], ResNet [195], Inception V3 [54,70] and many other models for classification. Fig. 12 depicts the number of articles that use various TL models. TL methods help boost the results of DL methods by reducing the training time, saving memory, and improving the accuracy [70,191]. However, they suffer from the issue of gradient exploding, and gradient vanishing. Short-cut connections can be used to help optimize DL networks to tackle the issue. The issue of exploding gradients can be prevented with certain changes made to the model architecture [161].

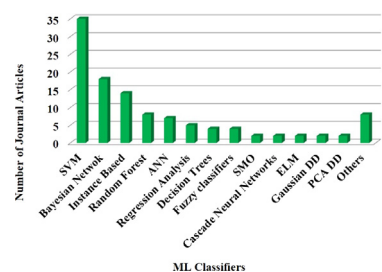


Fig. 10 – Depiction of number of articles using various ML classifiers for DME diagnosis.

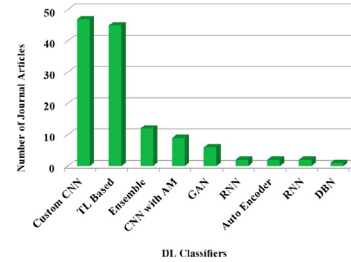


Fig. 11 – Number of state-of-art papers employing various DL models used for DME diagnosis.

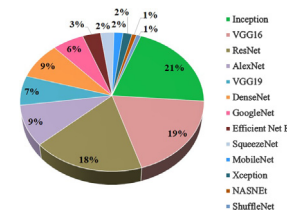


Fig. 12 – Percentage of state-of-art papers employing various TL models used for DME diagnosis.

The use of an appropriate activation function can aid in the prevention of vanishing gradient problems [166].

Simultaneously, few researchers had exhibited an interest in creating custom CNNs by training the models from scratch using very large datasets of the fundus and OCT images [161–163,173,187]. DL with CNN has several intriguing benefits which include application-specific feature learning, power of prediction, and generalization with superior results [224]. The use of CNNs for DME detection has shown tremendous results concerning accuracy [70,202,213,219,227]. Capsule networks helped incorporate positional information and hence provided accurate results (Accuracy-99.6%) for OCT classification [182]. Segmentation results obtained with an FCN model are far more accurate than those obtained with morphological techniques (mean intersection over union (meanIU)= 93.25 for FCN and meanIU = 71.88 for ML method) [176]. Subsequent research in the domain focused on the use of U-net-based models for image segmentation and classification [108,212,232,247]. Few TL models were used as mere feature extractors along with meta-heuristic wrapper methods used for feature selection. These extracted features were fed to ML-based classifiers [102].

Further research in the domain focused on the use of ensembles of ML-based, DL-based, and a blend of these two models to improve classification accuracy [85,97,224,228,232]. Ensemble learning is a method for strengthening the classification performance of a model by blending the results of numerous classifiers [69,85,227]. Ensemble models are effective and help avoid overfitting, but are relatively complex in their structure [222]. A model with an ensemble of CNNs can enhance the results (AUC-0.998) [69] compared to feature-based methods (AUC-0.903) [66].

Recent advances in the research focused on improving the performance of the encoder-decoder model of a U-net archi-

texture by incorporating the use of attention mechanisms [110,237]. In the recent past, GAN-based models, Object detection-based models, and AE-based models have been adopted, all of which are capable of outperforming previous techniques in terms of classification accuracy, interpretability, and model generalizability.

The key concern behind DL is that it demands a massive data to train the models. Data augmentation methods such as flipping, rotation, and translation can be used to improve the size of the training dataset [85,103,167]. With the use of GANs [98,99,248] and TL methods [70,111], weakly supervised learning can be useful in the case of limited data. Class imbalance issues are a common concern in DL which can be eliminated with the help of appropriate use of loss functions and data augmentation [85,233]. Over-fitting can be avoided by using data augmentation and model hyperparametrization techniques [167,191]. To counteract over-fitting, dropout can be used at every skip connection [164]. The fusion of hand-crafted features with the features extracted by DL methods helped in the accurate detection of disease [163]. Choosing the right hyperparameters for the model to improve the results is a vital part of DL [163,183]. Interpretability of data is an issue in DL as it is considered a “black box”. This problem can be alleviated to a certain level by generating heat-maps and visualization modules for better understanding prediction [20,149,212].

3.4.1. Future research directions

Most of the authors did not incorporate efficient preprocessing techniques which help reduce variability and noise present in OCT images. Low contrast, poor quality images were discarded [119,195]. There is a need to develop good preprocessing algorithms for retinal images to reduce the influence of artifacts. The presence of multiple pathologies in the single B scan makes it difficult to identify the disease accurately. Layer segmentation algorithms used for the DME pathology detection, were not efficient [124,147,149]. Efficient DL-based OCT layer and fluid segmentation algorithms are considered essential to help doctors simplify their tasks. While CFP is used to inspect lesions on the surface of the retina, OCT is being used to assess and detect fluid accumulation statistics as well as DME biomarkers. For better investigation, robust DME detection models can be constructed by taking into account the features retrieved from both the fundus and OCT images adopting DL [224]. The inspection of a single B scan from the OCT volume often is not sufficient for DME detection. For an accurate evaluation, it is always prudent to evaluate the entire OCT volume by including the evaluation of all B scans [138,239,240]. DME detection systems that can work with images acquired by multiple vendors could boost the generalizability of the model [167,215,247]. The inadequacy of DME screening methods to test people of diverse races is a concern on which further research is needed. The notion that an algorithm must be validated on broader patient cohorts in various settings, is a significant barrier to its adoption. Efficient DL models need to be designed for the detection and quantification of all three edema types even when they appear simultaneously within OCT images. Another major requirement in the area of DME detection stems from the fact that there are no public OCT

datasets available with annotations for three edema types and various DME biomarkers. The present state-of-the-art strategies have been validated with various datasets, all with no specifications regarding the selected circumstances used in the research, and under variable conditions [144,146,213].

Most of the DME detection methods used public OCT datasets with multiple eye disorders such as AMD, CNV, and DME. Most importantly, the detection of OCT biomarkers such as DRIL and HRF, disruption of EZ, and ELM can be combined with DME detection models to provide a comprehensive picture of the problem. Ophthalmologists will benefit immensely from a standalone DME detection system that can recognize multiple DME biomarkers, edema types, and their quantification. Although DL has shown to be effective in the field of medical image recognition, it is worth noting that it still has a few intrinsic limitations. Further research is needed to address the scarcity of high-quality labeled images, challenges of class imbalance, higher computational costs, and inadequate generalization ability.

4. Conclusion

Automated solutions provide a safe and convenient way to assist researchers and ophthalmologists around the world. This review provides a comprehensive analysis of the state-of-art computerized DME detection methods that use two imaging modalities fundus and OCT. This paper outlines all the fundus and OCT datasets that are publicly available, and the advantages and limitations of the CFP and OCT modalities along with ML-based and DL-based techniques used for DME detection. In addition, subcategories for ML-based and DL-based approaches are outlined, along with strategies for layer and fluid segmentation in OCT. Critical elements of future research are highlighted, along with a clear description of why they are essential. This review is unique since it covers a broad range of topics related to automated DME detection, as well as an overview of the dynamics of the advancements of the presented methods. The integration of OCT imaging techniques with effective DL algorithms could be a potential solution for the early detection of DME along with its assessment. Further research is appreciated for the identification and quantification of DME along with morphological variants, and OCT biomarkers to help monitor DME, aiding innovation in therapeutics.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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